



EdgeIPS™ LE

Administrator's Guide

V1.1
2022-09-26

Copyright © 2022 Trend Micro Incorporated. All rights reserved. No part of this publication may be reproduced, photocopied, stored in a retrieval system, or transmitted without the express prior written consent of Trend Micro Incorporated.



Trend Micro Incorporated reserves the right to make changes to this document and to the product described herein without notice. Before installing and using the product, review the readme files, release notes, and/or the latest version of the applicable documentation, which are available from the Trend Micro website at:

<http://docs.trendmicro.com/en-us/home.aspx>

Trend Micro, the Trend Micro t-ball logo, and TXOne Networks are trademarks or registered trademarks of Trend Micro Incorporated. All other product or company names may be trademarks or registered trademarks of their owners.

This documentation introduces the main features of the product and/or provides installation instructions for a production environment. Read through the documentation before installing or using the product.

Detailed information about how to use specific features within the product may be available at the Trend Micro Online Help Center and/or the Trend Micro Knowledge Base.

Table of Contents

Table of Contents.....	3
Chapter 1	6
About EdgeIPS [™] LE	6
Introduction	6
Main Functions.....	7
Extensive Support for Industrial Protocols	7
Policy Enforcement for Mission-Critical Machines.....	7
Improve Shadow OT Visibility by Integrating IT and OT Networks.....	7
Intrusion Prevention and Intrusion Detection	7
Switch Between Two Flexible Modes, ‘Monitor’ & ‘Prevention’	7
Top Threat Intelligence and Analytics	7
Centralized Management	7
Chapter 2	9
Getting Started.....	9
Getting Started: Task List	9
Opening the Management Console.....	10
Changing the Administrator’s Password.....	11
Chapter 3	12
The System Tab	12
Network Information & Device Information.....	12
Security Service Status & Throughput/Connection.....	12
Resource Monitor	13
Bandwidth Utilization	13
Packet Transmission Status	13
Chapter 4	14
The Visibility Tab	14
Active Query	14
Viewing Asset Information.....	14
Viewing Real Time Network Application Traffic.....	15
Chapter 5	16
The Device Tab.....	16
Configuring Device Settings	16
Configuring Interface Link Mode for Ports.....	17
Configuring LLDP Setting	17
Chapter 6	18
The Object Profiles Tab	18
Configuring IP Object Profiles	19
Configuring Service Object Profiles.....	19
Configuring Protocol Filter Profiles	20
Specifying Commands Allowed for a Protocol.....	21
Advanced Settings for SMB	21
Configuring IPS Profiles	22
Configuring a Pattern Rule for Granular Control.....	23
Configuring File Filter Profiles	25
Chapter 7	27
The Security Tab.....	27
Security General Settings	27
Inline Mode.....	27
Offline Mode.....	28
Configuring Security Operation Mode	28
Cyber Security	29
Configuring Cyber Security – Denial of Service Prevention	29
Policy Enforcement	30
Configuring Policy Enforcement	31
Adding Policy Enforcement Rules.....	32
Managing Policy Enforcement Rules.....	34
Chapter 8	37

The Pattern Tab.....	37
Viewing Device Pattern Information	37
Manually Updating the Pattern.....	37
Downloading Release Info	38
Chapter 9	39
The Application Tab	39
USB Application.....	39
Advanced USB Application	39
Packet Capture.....	40
Enabling Packet Capture.....	40
Download Captured Packet.....	40
Chapter 9	42
The Logs Tab.....	42
Viewing Cyber Security Logs.....	42
Viewing Policy Enforcement Logs	43
Viewing Protocol Filter Logs	43
Viewing File Filter Logs.....	44
Viewing Suspicious Object Logs.....	45
Viewing Asset Detection Logs	45
Viewing System Logs.....	46
Viewing Audit Logs.....	46
Chapter 10	47
The Administration Tab	47
Account Management.....	47
User Roles.....	47
Built-in User Accounts.....	48
Adding a User Account	48
Changing Your Password	48
Configuring Password Policy Settings.....	49
Auth Service	49
Configuring TACACS+	49
The System Management Tab	50
Configuring Device Name and Device Location Information	51
Configuring Management Method and Access Control List.....	51
Configuring Management Protocols and Ports	51
Configuring Control List Access from Management Clients	51
The Sync Settings Tab	52
Enabling Management by ODC.....	52
Action for Master Rule and Profile.....	52
The Syslog Tab	53
Configuring Syslog Settings	53
Syslog Severity Levels.....	54
Syslog Severity Level Mapping Table	54
The SNMP Tab	55
Configuring SNMP v1/v2c	55
Configuring SNMP v3	56
Configuring SNMP Trap Receivers	56
The System Time Tab.....	58
Configuring System Time	58
The Back Up / Restore Tab.....	59
Backing Up a Configuration	59
Restoring a Configuration.....	59
The Firmware Management Tab.....	60
Viewing Device Firmware Information	60
Upgrading Firmware.....	60
Rebooting and Applying Firmware.....	61
The Reboot System Tab.....	62
Rebooting the System	62
Chapter 11	63



Supported USB Devices	63
Supported actions via USB Disk	63
On-demand Configuration backup	65
Load Pattern from Disk	65
Load Configuration from Disk	66
Load Firmware from Disk	66
Appendix A.....	68
Terms and Acronyms.....	68

About EdgeIPS™ LE

Introduction

EdgeIPS™ LE is a palm-sized platform that is fitted with dual Ethernet LAN ports. Users can access its web-based management console that provides a graphical user interface for device configuration and security policy settings. The whole management process is designed to comply with the manufacturing SOPs of the industry. The EdgeIPS™ LE protects your individual assets with OT visibility, cybersecurity, and OT protocol allow-listing/deny-listing.

IT and OT traditionally are operated separately, each with its own network, transportation team, goals, and needs. In addition, each industrial environment is equipped with tools and devices that were not designed to connect to a corporate network, thus provisioning timely security updates or patches can be difficult. Therefore, the need for security products that provide proper security protection and visibility is on the rise.

Trend Micro provides a wide range of security products that cover both your IT and OT layers. These easy-to-build solutions provide an active and immediate protection to Industrial Control System (ICS) environments with the following features:

- Certified commercial grade hardware with size, power consumption, and durability tailored for OT environments.
- Threat detection and interception, with safeguards against the spread of worms
- Protection against Advanced Persistent Threats (APTs) and Denial of Service (DoS) attacks that target vulnerable legacy devices
- Virtual patch protection against OT device exploits

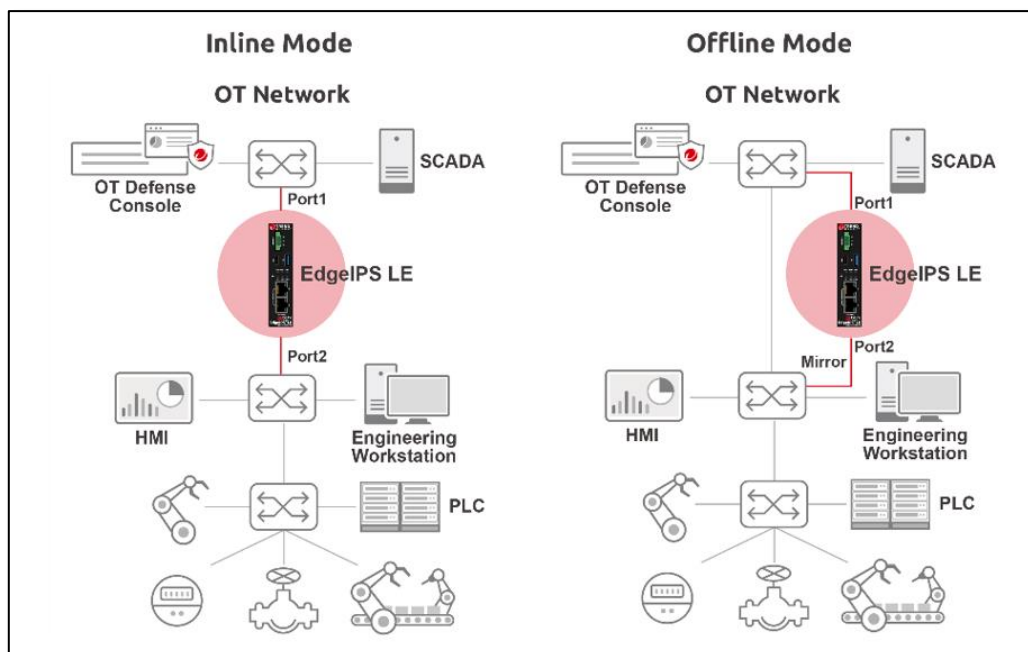


Figure 1. TXOne Networks security solutions for an OT network

Main Functions

EdgeIPS™ LE is a transparent network security device. The main functions of the product are as follows:

Extensive Support for Industrial Protocols

EdgeIPS™ LE supports the identification of a wide range of industrial control protocols, including Modbus and other protocols used by well-known international companies such as Siemens, Mitsubishi, Schneider Electric, ABB, Rockwell, Omron, and Emerson. In addition to allowing OT and IT security system administrators to work together, this feature also allows the flexibility to deploy defense measures in appropriate network segments and seamlessly connects them to existing factory networks.

Policy Enforcement for Mission-Critical Machines

EdgeIPS™ LE's core technology TXODI allows administrators to maintain a policy enforcement database. By analyzing Layer 3 to Layer 7 network traffic between mission-critical production machines, policy enforcement executes filtering of control commands within the protocols and blocks traffic that is not defined in the policy rules. This feature can help prevent unexpected operational traffic, block unknown network attacks, and block other activities that match the defined policy rules.

Improve Shadow OT Visibility by Integrating IT and OT Networks

EdgeIPS™ LE comes equipped to make your IT and OT networks as integrated and coordinated with each other as possible, and to grant visibility of your shadow OT environment.

Intrusion Prevention and Intrusion Detection

IPS/IDS provides a powerful, up-to-date first line of defense against known threats. Vulnerability filtering rules provide effective protection against exploits at the network level. Manufacturing personnel manage patching and updating, providing pre-emptive protection against critical production failures and additional protection for old or terminated software.

Switch Between Two Flexible Modes, 'Monitor' & 'Prevention'

EdgeIPS™ LE flexibly switches between 'Monitor' and 'Prevention' modes. In 'Monitor' mode the device will log traffic without interfering, while in 'Prevention' mode it will filter traffic based on policies you create. These modes work together to preserve your productivity while maximizing security.

Top Threat Intelligence and Analytics

EdgeIPS™ LE provides advanced protection against unknown threats with its up-to-date threat information. With the help of the Zero Day Initiative (ZDI) vulnerability reward program, EdgeIPS™ LE offers your systems exclusive protection from undisclosed and zero-day threats.

Centralized Management

TXOne's OT Defense Console (ODC) provides a graphical user interface for policy management in compliance with manufacturing SOP. It centrally monitors operations information, edits network protection policies, and sets patterns for attack behaviors.



All protections are deployed throughout the entire information technology (IT) and operational technology (OT) infrastructure. These include:

- A centralized policy deployment and reporting system
- Full visibility into assets, operations, and security threats
- IPS and policy enforcement configuration can be assigned per device group, allowing all devices in the same device group to share the same policy configuration
- Management permissions for device groups can be assigned per user account

Getting Started

This chapter describes the EdgeIPS™ LE and how to get started with configuring the initial settings.

Note: For an overview of the physical hardware and characteristics, or a more condensed manual to help with initial setup of the device, please refer to the document "EdgeIPS LE - Quick Setup Guide"

Getting Started: Task List

This task list provides a high-level overview of all procedures required to get EdgeIPS™ LE up and running as quickly as possible. Each step links to more detailed instructions found later in the document.

Procedure

1. Open the management console.
For more information, see [Opening the Management Console on page 10](#).
2. Change the administrator password.
For more information, see [Changing the Administrator's Password on page 11](#).
3. Configure the system time.
For more information, see [Configuring System Time on page 58](#).
4. (Optional) Configure the Syslog settings.
For more information, see [Configuring Syslog Settings on page 53](#).
5. Configure Object Profiles.
For more information, see [The Object Profiles on page 18](#).
6. Configure security policies.
For more information, see [The Security Tab on page 27](#).
7. Configure the device name and device location information.
For more information, see [Configuring Device Name and Device Location Information on page 51](#).
8. (Optional) Configure access control list from management clients.
For more information, see [Configuring Control List Access from Management Clients on page 51](#).
9. Configure management protocols and ports.
For more information, see [Configuring Management Protocols and Ports on page 51](#).
10. (Optional) Update the DPI (Deep Packet Inspection) pattern for the device.
For more information, see [Manually Updating the Pattern on page 37](#).
11. (Optional) Enabling Management by ODC.
For more information, see [Enabling Management by ODC on page 52](#).
12. Configure the network settings and network interface link modes for the device.
For more information, see [The Device Tab on page 16](#).

Opening the Management Console

EdgeIPS™ LE provides a built-in management web console that you can use to configure and manage the product. View the management console using a web browser.

Note: View the management console using Google Chrome version 63 or later; Firefox version 53 or later; Safari version 10.1 or later; Edge version 15 or later.

Procedure

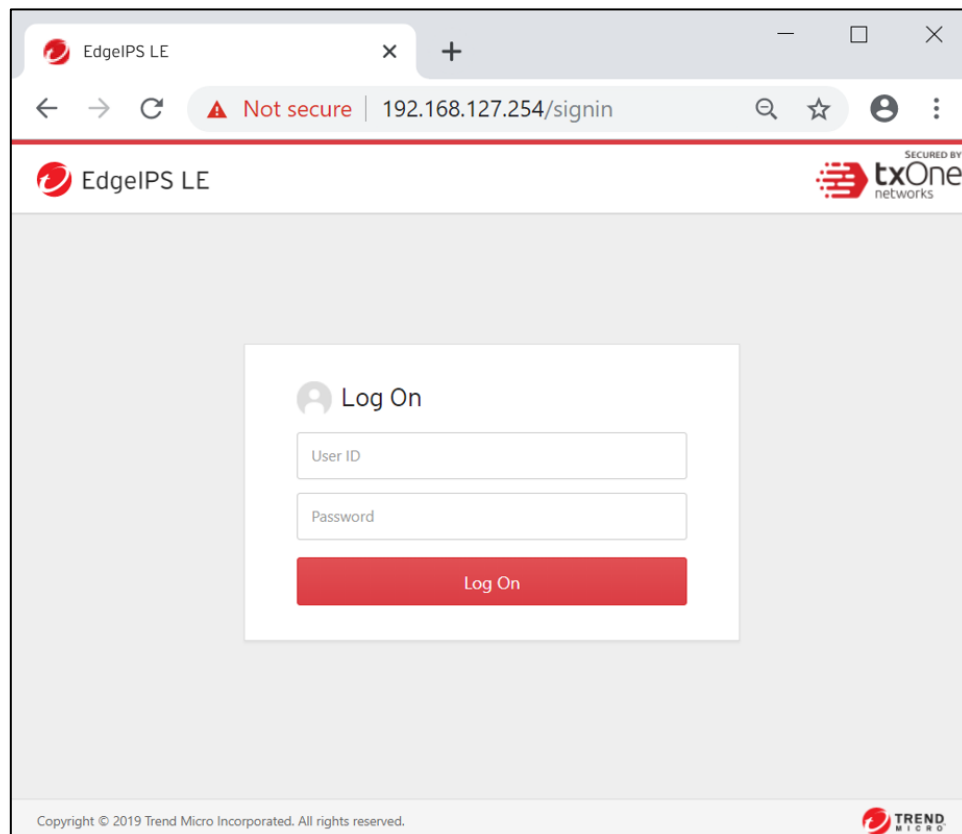
1. In a web browser, type the address of the EdgeIPS™ LE in the following format:

<https://192.168.127.254>

Note: The TXONE device uses an automatically generated self-signed SSL certificate to encrypt communications to and from the client accessing the device. Given that the certificate is self-signed, most browsers will not trust the certificate and will give a warning that the certificate being used is not signed by a known authority.

The logon screen will appear.

Note: The default IP address of EdgeIPS™ LE is 192.168.127.254 with subnet 255.255.255.0. Before you connect a PC/Laptop to EdgeIPS™ LE, the PC's IP address should be set to an IP address that is able to access the default IP address. After that, connect the PC and EdgeIPS™ LE using an Ethernet cable.



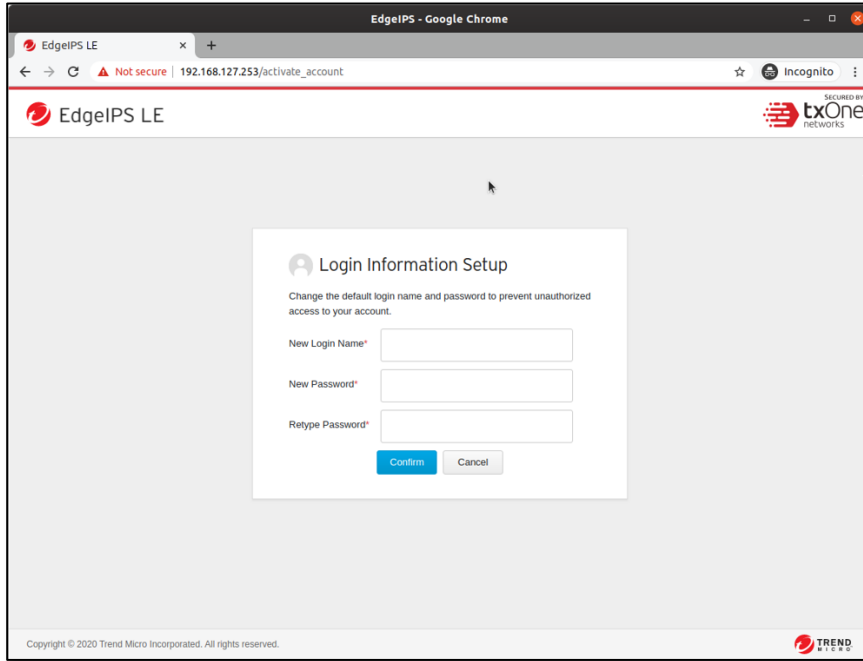
2. Input the logon credentials (user ID and password).

Use the default administrator logon credentials when logging on for the first time:

- User ID: admin
- Password: txone

3. Click Log On.

4. When logging in for the first time or after a factory reset, you will be prompted to change the default user ID and password. The default user ID and password cannot be used.



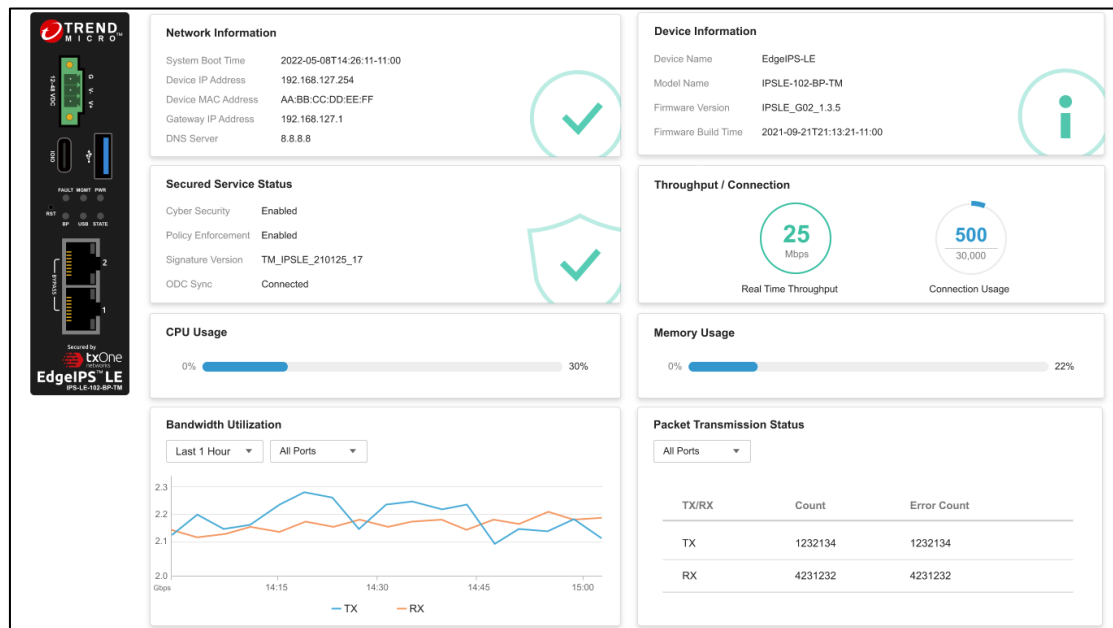
5. Log in with newly changed user ID/password credentials.

Changing the Administrator's Password

Refer to section [Changing Your Password](#) under Chapter 10 *The Administration Tab* > *Account Management*.

The System Tab

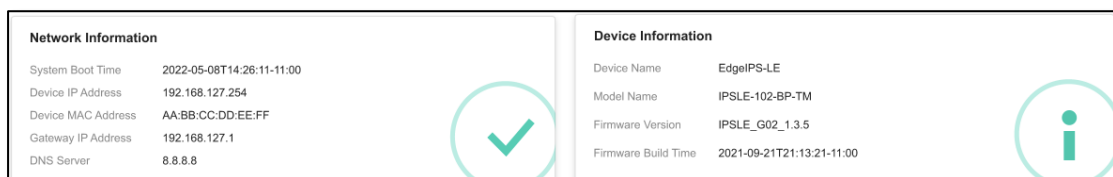
Monitor your system information, system status, and system resource usage on the System tab promptly.



Network Information & Device Information

The widget on the left shows network-related details, including system boot time, device IP address, device MAC address, Gateway IP address and DNS server.

The widget on the right shows device-oriented details, including device name, model name, firmware version of the device and firmware build date/time.

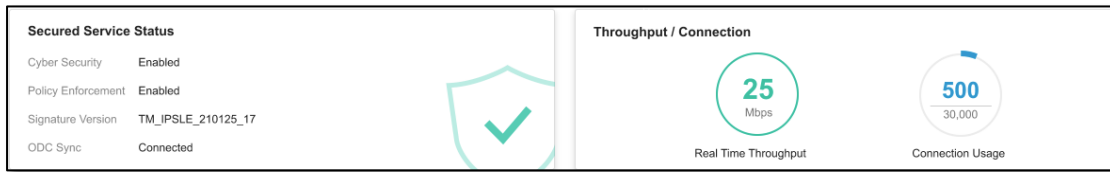


Security Service Status & Throughput/Connection

The widget on the left shows secured service status, including whether Cyber Security function is enabled, whether Policy Enforcement function is enabled, signature version of the device and whether the device is managed by the ODC.

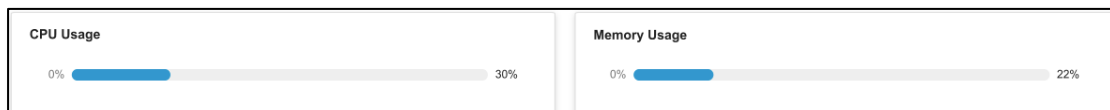
The widget on the right shows two types of visualized data. One is the current network throughput of the device, indicating the amount of data passing through a system; and the other is the current network connection usage of the device, indicating the utilization of

connection (according to the refresh time settings).



Resource Monitor

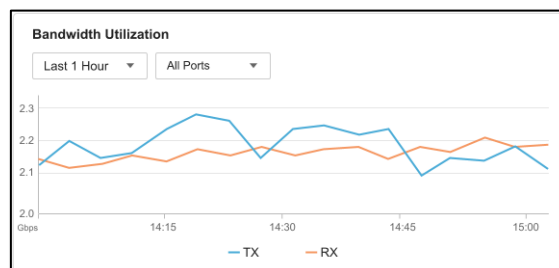
The two widgets show resource usages in terms of CPU and memory respectively.



Items	Descriptions
CPU Utilization	Real time CPU utilization % (according to the refresh time settings)
Memory Utilization	Real time memory utilization % (according to the refresh time settings)

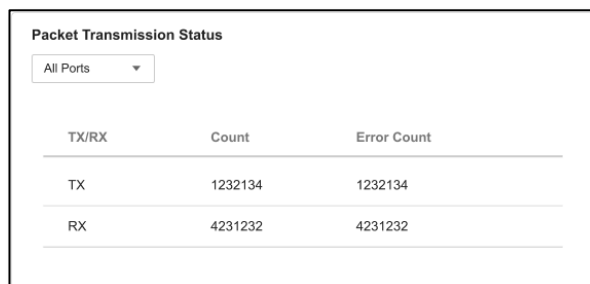
Bandwidth Utilization

This widget shows bandwidth utilization according to different time intervals by port status and shows TX and RX bandwidth utilization.



Packet Transmission Status

This widget shows packet transmission status by port, including interface TX/RX, count number, and error count info.



Packet Transmission Status

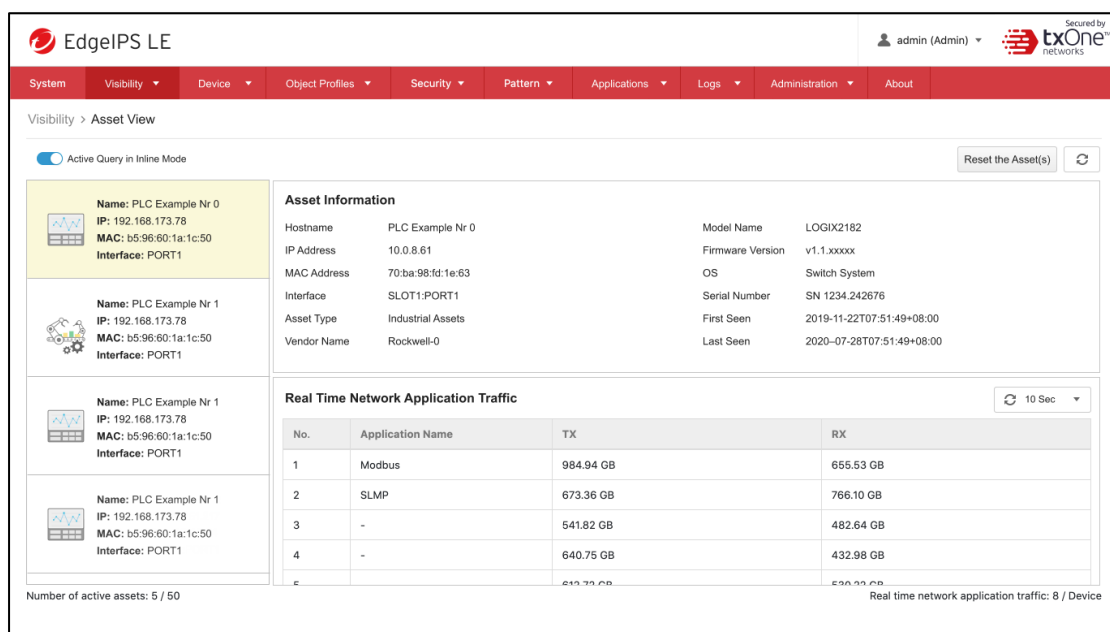
All Ports

TX/RX	Count	Error Count
TX	1232134	1232134
RX	4231232	4231232

The Visibility Tab

The Visibility screen gives you an overview of asset visibility for your managed assets. The screen provides you with timely and accurate information on the assets that are managed by EdgeIPS™ LE.

The assets, listed under the tab, are automatically detected by EdgeIPS™ LE devices.



EdgeIPS LE admin (Admin) txOne networks

System Visibility Device Object Profiles Security Pattern Applications Logs Administration About

Visibility > Asset View

Active Query in Inline Mode Reset the Asset(s)

Asset Information

Hostname	PLC Example Nr 0	Model Name	LOGIX2182
IP Address	10.0.8.61	Firmware Version	v1.1.xxxxx
MAC Address	70:ba:98:fd:1e:63	OS	Switch System
Interface	SLOT1:PORT1	Serial Number	SN 1234.242676
Asset Type	Industrial Assets	First Seen	2019-11-22T07:51:49+08:00
Vendor Name	Rockwell-0	Last Seen	2020-07-28T07:51:49+08:00

Real Time Network Application Traffic 10 Sec

No.	Application Name	TX	RX
1	Modbus	984.94 GB	655.53 GB
2	SLMP	673.36 GB	766.10 GB
3	-	541.82 GB	482.64 GB
4	-	640.75 GB	432.98 GB
5	-	643.33 GB	590.33 GB

Number of active assets: 5 / 50 Real time network application traffic: 8 / Device

Note: The term **asset** in this chapter refers to the devices or hosts that are protected by EdgeIPS™ LE.

Active Query

Active query can detect inactive or dormant assets or passive assets on the network.

Note: Active query operates only in Inline Mode. In Offline Mode, the switch button of active query is not configurable. In firmware 1.0, active query supports 4 protocols (Modbus, CIP, FINS and SMB).

Viewing Asset Information

Procedure

1. Go to [Visibility] > [Assets View].
2. Click an asset icon and view its detailed information.

	Name: PLC Example Nr 0 IP Addr: 192.168.173.78 MAC: b5:96:60:1a:1c:50 Interface: Port1
	Name: PLC Example Nr 1 IP Addr: 192.168.220.63 MAC: 90:8d:ce:da:71:db Interface: Port1
	Name: PLC Example Nr 2 IP Addr: 192.168.251.217 MAC: b3:80:da:10:46:74 Interface: Port1
	Name: PLC Example Nr 2 IP Addr: 192.168.251.217 MAC: b3:80:da:10:46:74 Interface: Port1

3. The [Assets Information] pane shows the following information for the asset:

Fields	Descriptions
Hostname	The hostname of the asset.
Model Name	The model name of the asset.
IP Address	The IP address of the asset.
Firmware Version	The firmware version of the asset.
MAC Address	The MAC address of the asset.
OS	The operating system of the asset.
Interface	The physical port of EdgeIPS™ LE that detects the connection with the asset.
Serial Number	The serial number of the asset.
Asset Type	The asset type of the asset.
First Seen	The date and time when the asset was first seen.
Vendor Name	The vendor name of the asset.
Last Seen	The date and time when the asset was last seen.

Viewing Real Time Network Application Traffic

Procedure

- Go to [Visibility] > [Assets View].
- Click an asset icon and view its detailed information.
- The [Real Time Network Application Traffic] pane shows a list of network traffic statics of the asset

Fields	Descriptions
No.	Ordinal number of the application traffic.
Application Name	The application type of the traffic.
TX (Kbytes / Mbytes)	The amount of traffic transmitted for this application.
RX (Kbytes / Mbytes)	The amount of traffic received for this application.

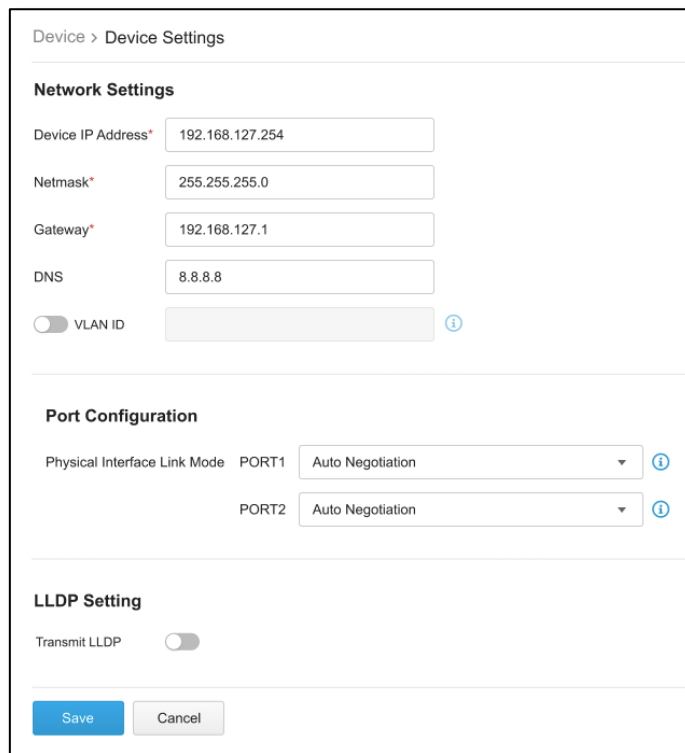
Note: Click the [Reset the Asset(s)] on the top-right of the window to set back to the initial state. This is only available for Admin role.

Note: Click the icon  to refresh the displayed information.

Specify the refresh interval under the [Refresh Time] dropdown menu.

The Device Tab

This chapter describes how to set up network settings and port configuration for the device.



The screenshot shows the 'Device > Device Settings' configuration page. It is divided into three main sections: Network Settings, Port Configuration, and LLDP Setting. The Network Settings section includes fields for Device IP Address (192.168.127.254), Netmask (255.255.255.0), Gateway (192.168.127.1), and DNS (8.8.8.8). There is also a toggle for 'VLAN ID' which is currently off. The Port Configuration section shows two ports, PORT1 and PORT2, both set to 'Auto Negotiation'. The LLDP Setting section has a 'Transmit LLDP' toggle which is also off. At the bottom, there are 'Save' and 'Cancel' buttons.

Device > Device Settings

Network Settings

Device IP Address* 192.168.127.254

Netmask* 255.255.255.0

Gateway* 192.168.127.1

DNS 8.8.8.8

☐ VLAN ID

Port Configuration

Physical Interface Link Mode PORT1 Auto Negotiation

PORT2 Auto Negotiation

LLDP Setting

Transmit LLDP ☐

Save Cancel

Configuring Device Settings

Procedure

1. Go to [Device] > [Device Settings]
2. In the [Network Settings] pane, configure the network settings for the device:
 - Device IP address
 - Netmask
 - Gateway
 - DNS
 - Enable VLAN ID
 - VLAN ID

Configuring Interface Link Mode for Ports

Procedure

1. Go to [Device] > [Device Settings].
2. In the [Port Configuration] pane, configure the link modes for the ports of the device:

Field	Description
Port 1 and Port 2	Choose [Auto Negotiation] to specify that the interface should automatically negotiate the highest speed that both sides can work with or specify the configured speed value of the interface. ☒ Auto Negotiation ☐ 10 Mbps Half Duplex ☐ 10 Mbps Full Duplex ☐ 100 Mbps Half Duplex ☐ 100 Mbps Full Duplex ☐ 1 Gbps Full Duplex

Configuring LLDP Setting

Procedure

1. Go to [Device] > [Device Settings].
2. In the [LLDP Setting] pane, configure the link modes for the ports of the device:

Field	Action
Transmit LLDP	Enable Transmit LLDP (Link Layer Discovery Protocol) for discovery and configuration to allow EdgeIPS™ LE to advertise its identity and capabilities on the network.

The Object Profiles Tab

Object profiles simplify policy management by storing configurations that can be used by EdgeIPS™ LE.

You can configure the following types of object profiles for this device:

- **IP Object Profiles:** Contain the IP addresses that you can apply to a policy rule.
- **Service Object Profiles:** Contain the service definitions that you can apply to a policy rule. TCP port range, UDP port range, ICMP, and custom protocol number are defined here.
- **Protocol Filter Profiles:** Contain basic protocol settings that you can apply to a policy rule. EdgeIPS™ LE supports general IT protocols only.
- **IPS Profiles:** Contain the settings of IPS (Intrusion Prevention System) pattern rules that you can create profiles or edit profiles to apply to a policy rule. Details of ICS (Industrial Control System) protocols are defined here.
- **File Filter Profiles:** Contain the settings of file filter profiles that you can apply to a policy rule. Details of file filter by protocol are defined here.

Each type of object profile has the following types of sub profile listed for this device.

- **Device Profiles:** Contain the IP Object Profile(s), Service Object Profile(s), Protocol Filter Profile(s), IPS Profile(s) and File Filter Profile(s) that users with local account privileges for EdgeIPS™ LE can apply to a policy rule.
- **Master Profiles:** Contain the IP Object(s), Service Object(s), Protocol Filter Profile(s), IPS Profile(s), and File Filter Profile(s) synced from ODC that users with local account privilege in ODC can apply to a policy rule. Users with local account privilege for EdgeIPS™ LE can only view and copy profiles(s) from Master Profiles(s) to Device Profiles.

The following table describes the tasks you can perform when you view a list of the profiles:

- **Device Profiles:**

Tasks	Descriptions
Add a profile	Click [Add] to create a new profile.
Edit a profile	Click a profile name to edit the settings.
Delete a profile	Select one or more profiles and click [Delete].
Copy a profile	Select on profile and click [Copy].

- **Master Profiles:**

Task	Description
Copy a profile	Select one or more profiles and click [Copy Master Profiles to Device Profiles].

Configuring IP Object Profiles

You can configure the IP address in an IP object profile, which can be used by other policy rules.

The types of IP address you can assign are:

- Single IP addresses
For example: 192.168.1.1
- IP ranges
For example: from 192.168.1.1 to 192.168.1.20
- IP subnets
For example: 192.168.1.0/24

Procedure

1. Go to [Object Profiles] > [IP Object Profiles].
2. Do one of the following:
 - Click [Add] to create a profile.
 - Click a profile name to edit settings.



3. Type a descriptive name for the Profile Name field.
4. Type a description.
5. Under the [IP Profile List], specify an IP address, an IP range, or an IP subnet.
6. If you want to add another entry, click the  button.
7. Click [OK].

Configuring Service Object Profiles

In a service object profile, you can define the following:

- TCP protocol port range
For example: TCP port 100 ~ 120
- UDP protocol port range
For example: UDP port 100 ~ 120
- ICMP protocol type and code
For example: ICMP type 8 code 0
- Custom protocol with specified protocol number
For example: protocol number = 6 and service ports range from 100 to 120

Note: The term 'protocol number' refers to the protocol number defined in the internet protocol suite.

Procedure

1. Go to [Object Profiles] > [Service Object Profiles].
2. Do one of the following:
 - Click [Add] to create a profile.
 - Click a profile name to edit settings.

Create Service Object Profile
✕

Profile Name*

i

Description

i

Service Object List

(Max: 8)

No.1*	TCP ▼	Protocol Number	6	Service Port	0	~	0	+
-------	---	-----------------	---	--------------	---	---	---	---

3. Type a descriptive name for the Service Object Profile.
4. Type a description.
5. Provide one of the following definitions:
 - a. TCP protocol and its port range
 - b. UDP protocol and its port range
 - c. ICMP protocol and its type and code
 - d. Custom protocol with specified protocol number
6. If you want to add another entry, click the + button.
7. Click [OK].

Configuring Protocol Filter Profiles

A protocol filter profile contains more sophisticated and advanced protocol settings that you can apply to a policy rule. After you update the signature, the new item of protocol will be shown for your configuration. There is no need to wait for new firmware update in order to get the new protocol support, which can shorten the waiting gap during POC stage and sales pitch.

The following can be configured in a protocol filter profile:

- | | |
|---|---|
| <ul style="list-style-type: none"> ▪ DNS ▪ FTP ▪ HTTP ▪ MQTT ▪ MSRPC ▪ RDP ▪ SIP | <ul style="list-style-type: none"> ▪ SMTP ▪ SNMP ▪ SSH ▪ TELNET ▪ TFTP ▪ VNC ▪ SMB |
|---|---|

Specifying Commands Allowed for a Protocol

When configuring a protocol, you can specify which commands will be included in the protocol profile, as the following picture shows.

Command / Function Category Access Permission


☒ Any
 ☐ Basic

☐ Read Only
 ☐ Read / Write
 ☐ Admin Config
 ☐ Others

Advanced Settings for SMB

The device features more detailed configurations for the SMB protocol. Through the [SMB Advanced Settings] column, you can specify the settings in more details.

SMB Advanced Settings ×

SMB Protocol
☒ SMB v1/v2/v3
☐ SMB v1
☐ SMB v2/v3

☐ File Access Permission
☒ Read / Write
☐ Read Only
☐ Write Only

☐ Drop Anonymous Access for SMB v1/v2/v3

OK

Cancel

Procedure

- Go to [Object Profiles] > [Protocol Filter Profiles].
 - Click [Add] to add a protocol filter profile.
- The [Create Protocol Filter Profile] screen will appear.

Create Protocol Filter Profile ×

Profile Name* ⓘ

Description ⓘ

Enabled: 0 | Disabled: 13 | Total: 13

All Statuses ▾
 All Protocols ▾

☐	Status	Vertical	Protocol	Advanced Settings
☐	☐	Common	FTP	-
☐	☐	Common	HTTP	-
☐	☐	Common	SMB	SMB: v1/v2/v3, File Access Permission: Disabled, Drop Anonymous Access for SMB... 
☐	☐	Common	DNS	-
☐	☐	Common	MQTT	-
☐	☐	Common	MS RPC	-
☐	☐	Common	RDP	-
☐	☐	Common	SIP	-

Records: 1-13 / 13
 25 ▾
 Per page
 1 / 1
 << < > >>

Save

Cancel

3. Type a profile name for the protocol filter.
4. Type a description.
5. Select the protocols you want to include in the protocol filter.
 - a. Click the enable button in the [Status] column.
 - b. Click the [/] icon in the [Advanced Settings] column, then select one of the following:
 - **SMB Protocol** - Specify SMB the protocol version combination – options include SMBv1/v2/v3, SMBv1 and SMB v2/v3.
 - **File Access** – Select access permission behavior:
 - **Read / Write:** Read and write file access
 - **Read Only:** File access for reading only
 - **Write Only:** File access for writing only
 - **Drop Anonymous Access for SMB v1/v2/v3:** Drop access over SMB v1/v2/v3 for Anonymous accounts.

Configuring IPS Profiles

An IPS profile contains more sophisticated pattern rules that you can do granular control and apply to a policy rule.

The following can be configured in an IPS profile:

- Details of IPS protocol category, including:
 - File Vulnerabilities
 - Buffer Overflow
 - DoS Attack
 - Exploits
 - Malware Traffic
 - Reconnaissance
 - Web Threats
 - ICS Threats
 - Others
 - Misc
- Details of IPS protocol risk level category, including:
 - Information
 - Low
 - Medium
 - High
 - Critical
- Details of default action list for IPS patterns, including:
 - Accept and Log
 - Deny and Log

Object Profiles > IPS Profiles

Device Profiles		Master Profiles	
+ Add Total Number of Records: 2 (Max: 32)			
☐	No.	Profile Name	Description
☐	1	IPS_Profile_1	
☐	2	IPS_Profile_2	

Records: 1-2 / 2 25 per page 1 / 1 << < > >>

Configuring a Pattern Rule for Granular Control

When configuring an IPS pattern rule protocol, you can specify which action should be taken and add it to the IPS profile, as the following picture shows.

Create IPS Profile

Profile Name* ⓘ

Description ⓘ

Enabled: 5000 | Disabled: 232 | Total: 5232

All Statuses All Categories All Risk Levels All Actions Search

☐	Status	Rule ID	Category	Risk Level	Actions	Rule Name
☐	☒	1137562	Buffer Overflow	High	Accept and Log	WEB-CLIENT Microsoft Internet Explorer Memory Corruption Vulnerability (CVE-2015-1626)
☐	☐	1130512	Web Threats	High	Deny and Log	WEB-CLIENT Microsoft Internet Explorer Elevation of Privilege Vulnerability (CVE-2015-0072)
☐	☒	1130511	Web Threats	High	Accept and Log	WEB-CLIENT Microsoft Internet Explorer JPEG XR Parser Information Disclosure Vulnerability (CVE-2015-0076)
☐	☒	1130510	Buffer Overflow	High	Deny and Log	WEB-CLIENT Microsoft Internet Explorer Memory Corruption Vulnerability (CVE-2015-1625)
☐	☒	1130513	Buffer Overflow	High	Accept and Log	WEB-CLIENT Microsoft Internet Explorer Memory Corruption Vulnerability (CVE-2015-1626)
☐	☐	1130512	Web Threats	High	Accept and Log	WEB-CLIENT Microsoft Internet Explorer Elevation of Privilege Vulnerability (CVE-2015-0072)
☐	☒	1130511	Web Threats	High	Deny and Log	WEB-CLIENT Microsoft Internet Explorer JPEG XR Parser Information Disclosure Vulnerability (CVE-2015-0076)
☐	☒	1130510	Buffer Overflow	High	Deny and Log	WEB-CLIENT Microsoft Internet Explorer Memory Corruption Vulnerability (CVE-2015-1625)
☐	☐	1130513	Buffer Overflow	High	Accept and Log	WEB-CLIENT Microsoft Internet Explorer Memory Corruption Vulnerability (CVE-2015-1626)

Records: 1-25 / 5232 25 Per page 1 / 210 << < > >>

Save Cancel

IPS Rule Details

Status ☒

Rule ID 1137562

Profile Name WEB-CLIENT Microsoft Internet Explorer Memory Corruption Vulnerability (CVE-2015-1626)

Category Buffer Overflow

Risk Level High

Impact Remote Code Execution

Actions ☒ Accept and Log ☐ Deny and Log

Reference CVE-2015-1626; MS 15-018

TID T1234.003;

Keyword -

Save Cancel

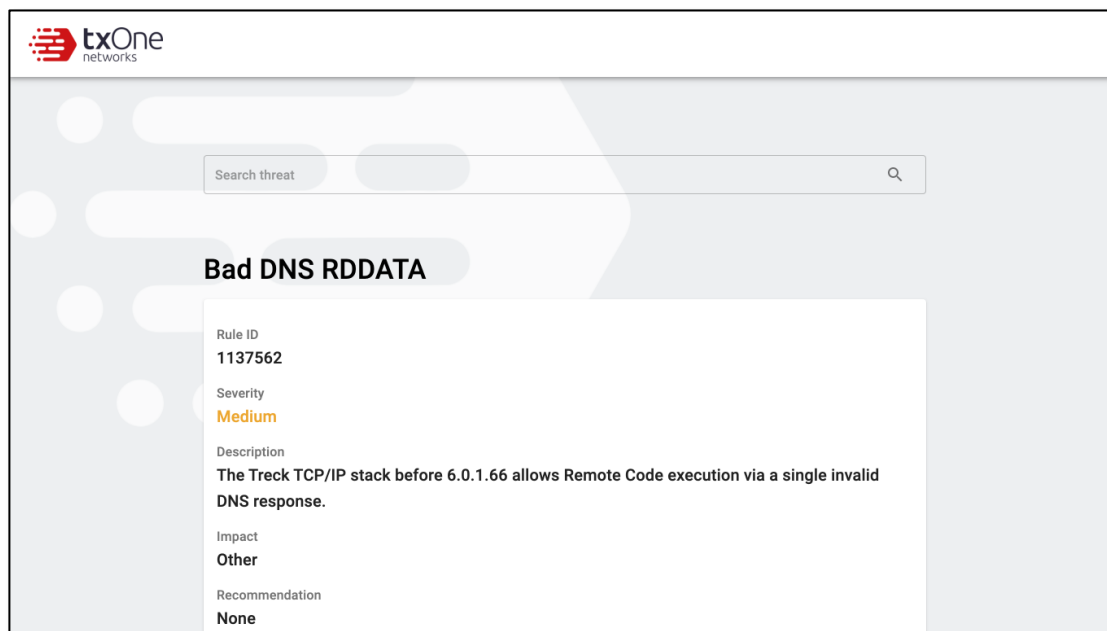
Procedure

1. Go to [Object Profiles] > [IPS Profiles].
2. Click [Add] to add an IPS profile.
The [Create IPS Profile] screen will appear.
3. Type a profile name for the IPS profile.

4. Type a description.
5. Select a pattern rule you want to configure by clicking on the rule ID.
6. IPS rule details will show up. Select one of the following:
 - **Status** - Specify the pattern rule to be enabled or disabled.
 - **Actions** - Multiple selections as follows:
 - **Accept and Log**: When an attack is detected by EdgeIPS™ LE, the attack will be bypassed and logged for monitoring.
 - **Deny and Log**: When an attack is detected by EdgeIPS™ LE, the attack will be blocked and logged for monitoring.

Fields	Descriptions
Status	The operational status of the pattern rule
Rule ID	The pattern rule ID
Rule Name	The rule name for the cyber attack
Category	The threat category of the cyber attack
Risk Level	The suggested security level of the cyber attack
Impact	The damage that will be caused to the target network device if the cyber attack succeeds
Actions	The preset actions for the cyber attack
Reference	The vulnerability ID of the cyber attack (e.g. CVE-2017-0147)
TID	MITRE ID information
Keyword	The word(s) for searching the pattern rules

7. If you want to see detailed threat information, please click Rule ID info and you will be redirect to TXOne Threat Encyclopedia website.



The screenshot shows the TXOne Threat Encyclopedia interface. At the top is the TXOne Networks logo. Below it is a search bar labeled 'Search threat'. The main content area displays the title 'Bad DNS RDDATA'. Below the title, there is a white box containing the following information:

- Rule ID: 1137562
- Severity: Medium
- Description: The Treck TCP/IP stack before 6.0.1.66 allows Remote Code execution via a single invalid DNS response.
- Impact: Other
- Recommendation: None

8. If you have already configured the pattern rule, press [Save].

Configuring File Filter Profiles

File Filter Profiles contain detailed access protocols, executable file types and Active Directory (AD) GPO dispatch settings, allowing you to create or edit profiles to apply to a policy rule to block files.

In a profile, you can define the following:

- **File Filter by Protocol**
Including: HTTP, FTP and SMB
- **Drop Executable Files**
Including: Windows (PE files) and Linux (ELF file)
- **Filter AD GPO**
Enable or disable Drop Active Directory (AD) GPO

Object Profiles > File Filter Profiles

Device Profiles						Master Profiles	
+ Add						Total Number of Records: 2 (Max: 32)	
☐	No.	Profile Name	File Filter Profile Settings	Filter AD GPO	Description		
☐	1	File_Filter_Profile_1	Enabled: HTTP, FTP, SMB	Enabled			
☐	2	File_Filter_Profile_2	Enabled: HTTP, FTP, SMB; Windows (PE File)	Disabled			

Records: 1-2 / 2 25 per page 1 / 1 << < > >>

Procedure

1. Go to [File Filter Profiles] > [File Filter Object Profiles].
2. Do one of the following:
 - Click [Add] to create a profile.
 - Click a profile name to edit settings.

Create File Filter Profile

Profile Name*

File_Filter_Profile_3

i

Description

i

File Filter Profile Settings

☐

File Filter by Protocol

Protocol*

☐ HTTP
 ☐ FTP
 ☐ SMB

Drop the Executable File(s)*

☐ Windows (PE File)
 ☐ Linux (ELF File)

Filter AD GPO

☐

Drop Active Directory (AD) GPO Dispatch

Save

Cancel

3. Type a descriptive name for the File Filter Profile Name field.
4. Type a description.
5. Under the [File Filter Profile Settings], enable file filter by protocol
 - File Filter by Protocol, including: HTTP, FTP and SMB.
 - Drop Transfer of Packed Executable Files, including: Windows (PE files) and Linux (ELF files)

6. If you want to filter AD GPOs, you can enable "Drop Active Directory (AD) GPO dispatch"
7. Click [Save] to save the profile.

Note: The support list of archive file type includes zip and gzip file type.

The Security Tab

This chapter describes security general settings, cyber security, policy enforcement, and suspicious objects.

Security General Settings

Use the [Security General Settings] tab to configure the security operation mode of the device.

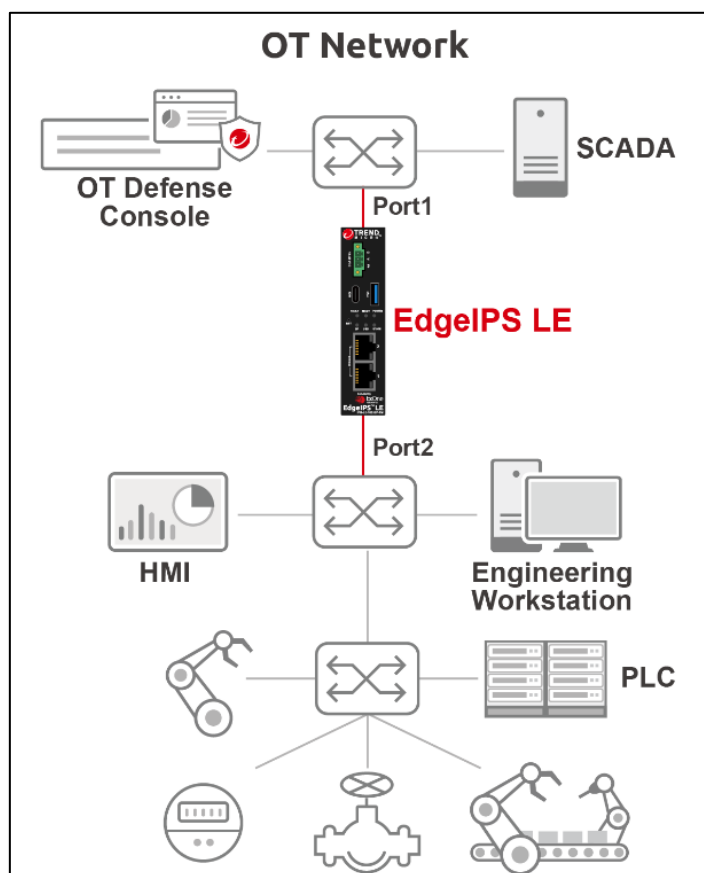
EdgeIPS™ LE offers two operation modes:

- **Inline Mode**
- **Offline Mode**

The following sections describe these two modes in detail.

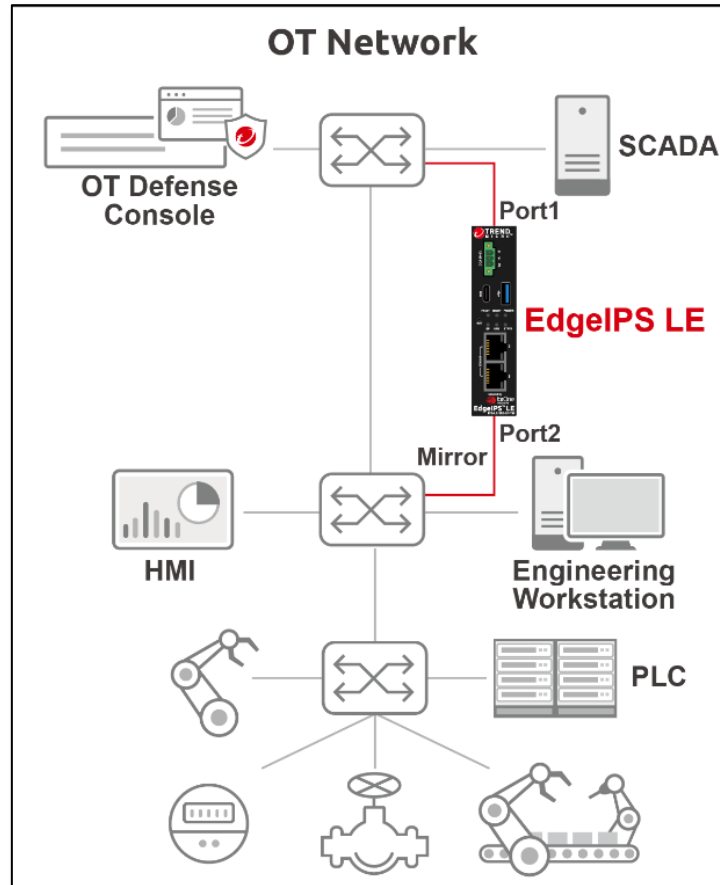
Inline Mode

EdgeIPS™ LE sits in the direct communication path between source and destination, actively analyzing, filtering, and taking actions on all traffic that passes through it.



Offline Mode

Data packets are mirrored from a core or other type of switch to port 2 of the EdgeIPS™ LE, which keeps detecting, monitoring, as well as outputting detection logs if threat events are detected.

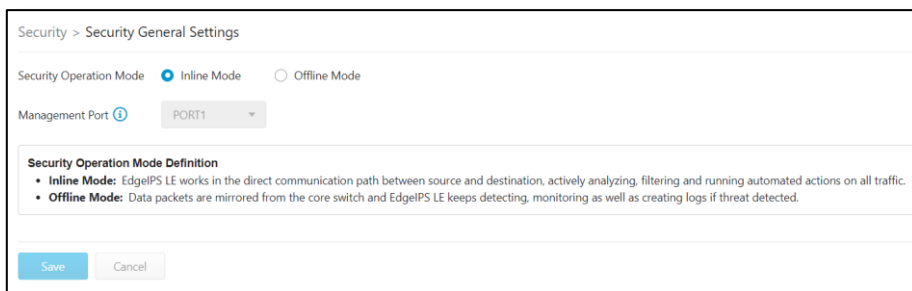


Note: Port 1 of the EdgeIPS™ LE functions serves as the management port, which connects to another switch, allowing the EdgeIPS™ LE to be managed by ODC.

Configuring Security Operation Mode

Procedure

1. Go to [Security] > [Security General Settings]
2. Under the [Security General Settings] tab you will see the following screen.



The screenshot shows the 'Security > Security General Settings' configuration screen. It includes the following elements:

- Security Operation Mode:** Two radio buttons, 'Inline Mode' (selected) and 'Offline Mode'.
- Management Port:** A dropdown menu showing 'PORT1'.
- Security Operation Mode Definition:**
 - Inline Mode:** EdgeIPS LE works in the direct communication path between source and destination, actively analyzing, filtering and running automated actions on all traffic.
 - Offline Mode:** Data packets are mirrored from the core switch and EdgeIPS LE keeps detecting, monitoring as well as creating logs if threat detected.
- Buttons:** 'Save' and 'Cancel' buttons at the bottom.

3. Choose a desired operation mode for this device.

Modes	Actions
Inline Mode	Choose [Inline Mode] to have EdgeIPS™ LE operate in Inline Mode. Configured IP setting in [Network Settings] pane can be connected from Port 1 or Port 2 at the same time.
Offline Mode	Choose [Offline Mode] to have EdgeIPS™ LE operate in Offline Mode. Configured IP setting in [Network Settings] pane can be connected from the selected physical port. The default port is Port1.

Note: EdgeIPS™ LE can log IT protocol activities from the mirror port of a switch if a protocol filter profile is configured and applied to the policy enforcement rule.

4. Choose the management port of the device if you select Offline Mode.

Note: When you switch from Inline Mode to Offline Mode for the first time, please note that you MUST connect to the physical port for device management in case you are unable to access the web console. After successfully switching to Inline Mode, you can specify Port 1 or Port 2 as the port to receive the traffic from the network device for monitoring and logging.

5. Click [Save].

Warning! Ensure that the operation mode is correctly selected. If EdgeIPS™ LE is deployed using inline network topology with the [Security Operation Mode] being set to [Offline Mode], then devices that connect to the non-management port cannot get through to outside.

Cyber Security

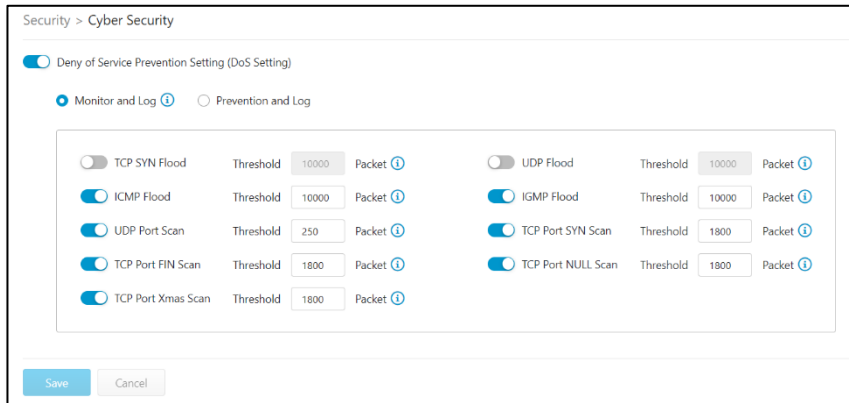
This device features cyber security, which covers denial of service attack prevention. The signature rules of intrusion prevention are called "TXOne One-pass Deep Packet Inspection for Industrial (TXODI™) Pattern." This pattern is provided by TXOne and can be regularly updated through ODC as well as by manual import via the device's web management UI.

Configuring Cyber Security – Denial of Service Prevention

Procedure

1. Go to [Security] > [Cyber Security].

- Under the [Cyber Security] tab you will see the [Denial of Service Prevention Setting (DoS Setting)] pane.



- Use the toggle to enable or disable the denial of service prevention feature.
- Select an action ([Monitor and Log] or [Prevention and Log]) for the feature.
- You can optionally configure the thresholds of the denial of service rules.
- Click [Save].

Note: Flood/Scan Attack Protection rules utilize detection period and threshold mechanisms to detect an attack. During a detection period (typically every 5 seconds), if the number of anomalous packets reaches the specified threshold, an attack detection occurs. If the rule action is [Block], the security node blocks subsequent anomalous packets until the end of the detection period. After the detection period, the security node will again allow anomalous packets until the threshold is reached.

The following table summarizes the settings:

IPS Operation Modes (Security General Setting)	Action Settings	Actions Performed
Inline Mode	Monitor and Log	▪ Detects and monitors network attacks, but does not block network attacks. ▪ Generates logs.
	Prevent and Log	▪ Blocks network attacks. ▪ Generates logs.
Offline Mode	Monitor and Log	▪ Passively detects and monitors network attacks. ▪ Generates logs.

Policy Enforcement

Policy enforcement allows you to define rules with various conditions, including the protocols for your own industry. Under which conditions with specified protocols, the activity will be allowed or blocked is based on to the policy you have set in the adopted ruleset template.

There are two types of policy enforcement rule list for this device:

- **Device Rule List:** Contains the policy rule(s) created and configured by users with local account privileges for EdgeIPS™ LE.
- **Master Rule List:** Contains the policy rule(s) synced from ODC that users who have local account privileges in ODC can apply to a policy rule. Users with local account

privileges for EdgeIPS™ LE can only view policy rules.

Note: For rule checking, the priority of device rule list is higher than the master rule list.

The following table describes the tasks you can perform when you view a list of the profiles:

- **Device Rule List:**

Tasks	Descriptions
Add a policy rule	Click [Add] to create a new policy rule.
Edit a policy rule	Click a policy rule name to edit the settings.
Delete a policy rule	Select one or multiple policy rules and click [Delete].
Copy a policy rule	Select a policy rule and click [Copy].
Change priority	Select a policy rule and click [Change Priority].

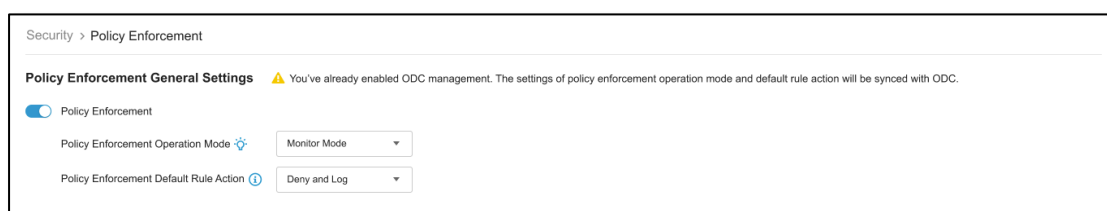
- **Master Rule List:**

Task	Description
N/A	The rule(s) on the Master Rule List are view-only.

Configuring Policy Enforcement

Procedure

1. Go to [Security] > [Policy Enforcement].
2. Under the [Policy Enforcement] tab, you will see the [Policy Enforcement General Settings] pane.



3. Use the toggle to enable or disable the policy enforcement feature.
4. Select a mode ([Monitor Mode], or [Prevention Mode]) for the feature.
5. In the [Policy Enforcement Default Rule Action] pane, select a default action [Accept], [Accept and Log] or [Deny and Log] when no policy enforcement rule is matched.

The following table summarizes the settings:

Security General Settings	Policy Enforcement Modes	Actions Performed
Inline Mode	Monitor Mode	▪ Detect and monitor abnormal protocol accesses to the OT assets, without blocking network attacks. ▪ Generate logs.
	Prevention Mode	▪ Block abnormal protocol accesses to OT assets. ▪ Generate logs.
Offline Mode	Monitor and Log	▪ Passively detects and monitors abnormal protocol accesses to OT assets. ▪ Generates logs.

Adding Policy Enforcement Rules

Procedure

1. Configure the required object(s).
 - IP Object Profiles
For more information, see [Configuring IP Object Profile on page 19](#).
 - Service Object Profiles
For more information, see [Configuring Service Object Profile on page 19](#).
 - Protocol Filter Profiles
For more information, see [Configuring Protocol Filter Profiles on page 20](#).
 - IPS Profiles
For more information, see [Configuring IPS Profiles on page 22](#).
 - File Filter Profiles
For more information, see [Configuring File Filter Profiles on page 25](#).
2. Go to [Security] > [Policy Enforcement].
3. Under the [Policy Enforcement] tab, you will see the following panes.

Policy Enforcement Rule List 

Device Rule List Master Rule List

[+ Add](#) Total Number of Records: 400 (Max: 512)

☐	No.	Status	Rule Name	Source IP / Object	Source IP / Object Info	Destination IP / Object	Destination IP / Object Info	Service Object	Service Info	VLAN ID	Action	Protocol Filter Profile	Protocol Filter Action	IPS Profile	File Filter Profile
☐	1		Rule-1	Any	Any	Any	Any	Any	Any	Any	Accept	Modbus ...	Deny	Disable	Disable
☐	2		Rule-2	Any	Any	Any	Any	Any	Any	1, 5, 10	Deny and...	-	-	-	-
☐	3		Rule-3	Any	Any	Any	Any	Any	Any	3, 20, 99	Deny and...	-	-	-	-
☐	4		Rule-4	Any	Any	Any	Any	Any	Any	3-15	Accept	CIP Prof...	Accept	Protocol ...	Disable
☐	5		Rule-5	Any	Any	Any	Any	Any	Any	100	Deny and...	-	-	-	-

4. Click the [Add] button to add a new policy rule.
5. Toggle to enable or disable the policy rule.

Create Policy Enforcement Rule 

Status 

Rule Name* 

Description 

Basic Filter

Source IP / Object

Destination IP / Object

Service Object

VLAN ID

Action

Advanced Filter 

☐ Protocol Filter Profile

Action

☐ IPS Profile

☐ File Filter Profile

OK

Cancel

6. Input a descriptive [Rule Name].
7. Input a descriptive [Description] for the rule.

8. On the pane of [Basic Filter], from the [Source IP / Object Profile] drop-down menu, select either one of the following for the source IP address(es):
- Any
 - Single IP
 - IP Range
 - IP Subnet
 - IP Object

Note: If you select [IP Object], then you need to select the IP object from an IP object profiles that have been created beforehand.

9. From the [Destination IP / Object Profile] drop-down menu, select either one of the following for the destination IP address(es):
- Any
 - Single IP
 - IP Range
 - IP Subnet
 - IP Object
10. From the [Service Object] drop-down menu, select either one of the following for the layer 4 criteria:
- Any
 - TCP
You can further specify the port range for this protocol.
 - UDP
You can further specify the port range for this protocol.
 - ICMP
You can further specify the Type and Code for this protocol.
 - Custom
You can further specify the protocol number for this protocol. The term protocol number refers to the one defined in the internet protocol suite.
 - Service Object

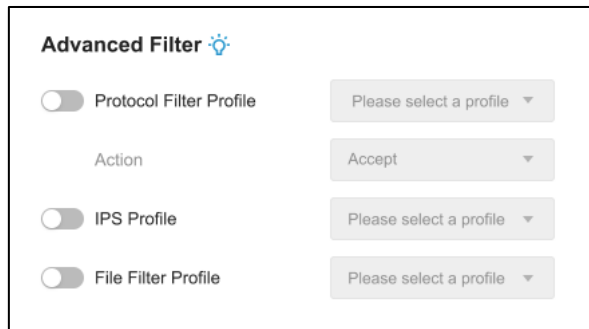
Note: You need to select the service object from service object profiles that have been created beforehand.

Enable [VLAN ID] button and Input the VLAN ID if needed.

Note: If you input the VLAN ID, the policy enforcement rule will ONLY check the traffic with the designated VLAN ID. If the VLAN ID field is blanked, the policy enforcement rule will check all the traffic whether it has a VLAN ID tag or not.

11. Under the [Action] drop-down menu, select one of the following:
- c. Accept: Select this option to allow network traffic that matches this rule.
 - d. Accept and Log: Select this option to allow network traffic that matches this rule and generate log(s).
 - e. Deny and Log: Select this option to block network traffic that matches this rule and generate log(s).
12. In the pane of [Advanced Filter], the following settings of profiles will be configured. The node will take further actions based on the protocol filter, the IPS profile or the File filter Profile.

- Under the [Protocol Filter Profile] drop-down menu, select a protocol filter profile you have defined beforehand. From the [Action] drop-down menu, select one of the following:
 - a. Accept: Select this option to allow network traffic that matches this rule.
 - b. Accept and Log: Select this option to allow network traffic that matches this rule and generate log(s).
 - c. Deny and Log: Select this option to block network traffic that matches this rule and generate log(s).
- Under the [IPS Profile] drop-down menu, select an IPS profile you have defined beforehand.
- Under the [File Filter Profile] drop-down menu, select a file filter profile you have defined beforehand.



Advanced Filter 

☐ Protocol Filter Profile Please select a profile ▼

Action Accept ▼

☐ IPS Profile Please select a profile ▼

☐ File Filter Profile Please select a profile ▼

13. Click [OK] to save the configuration.

Managing Policy Enforcement Rules

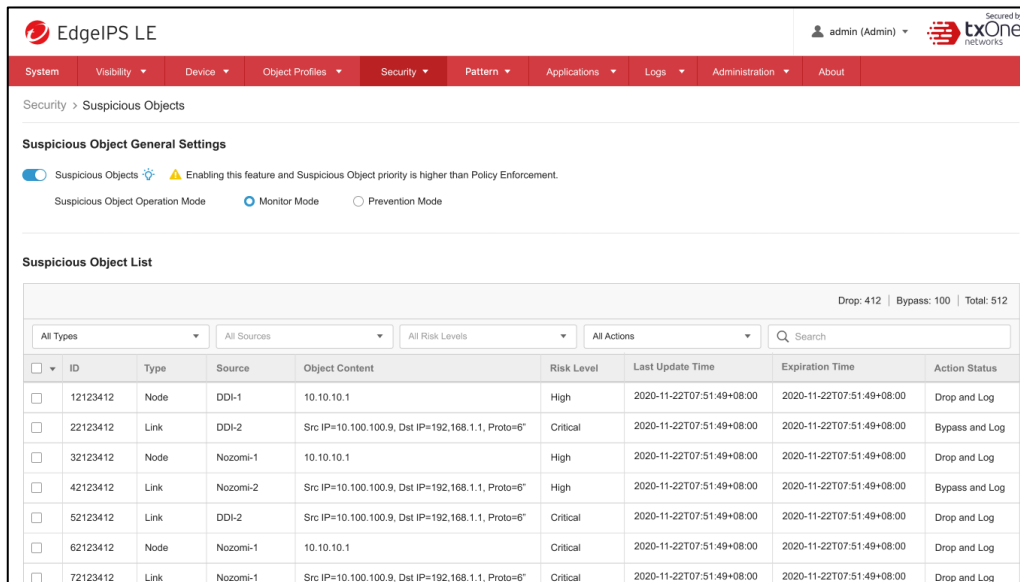
The following table lists common tasks that are used to manage policy enforcement rules.

Tasks	Actions
To delete a policy enforcement rule	Click the check box in front of a policy enforcement rule and click the [Delete] button.
To duplicate a policy enforcement rule	Click the check box in front of a policy enforcement rule and click the [Copy] button.
To edit a policy enforcement rule	Click the name of a rule, and an [Edit Policy Enforcement Rule] window will appear.
To change the priority of a policy enforcement rule	Click the check box in front of a policy enforcement rule, click the [Change Priority] button, and specify a new priority for this rule.

Note: When more than one policy enforcement rule is matched, EdgeIPS™ LE takes the action defined in the rule with the highest priority and ignores the rest of the rules. The rules are listed on the table of the UI screen by priority, with the highest priority rule listed on the first row of the table.

Suspicious Objects

Suspicious Objects allow you to define a custom node-based/link-based list that matches to the hash value of a network node or a network link, and then allow or block activities fitting that network node/network link in your network environment.



Suspicious Object General Settings

☒ Suspicious Objects Enabling this feature and Suspicious Object priority is higher than Policy Enforcement.

Suspicious Object Operation Mode: ☒ Monitor Mode ☐ Prevention Mode

Suspicious Object List

Drop: 412 | Bypass: 100 | Total: 512

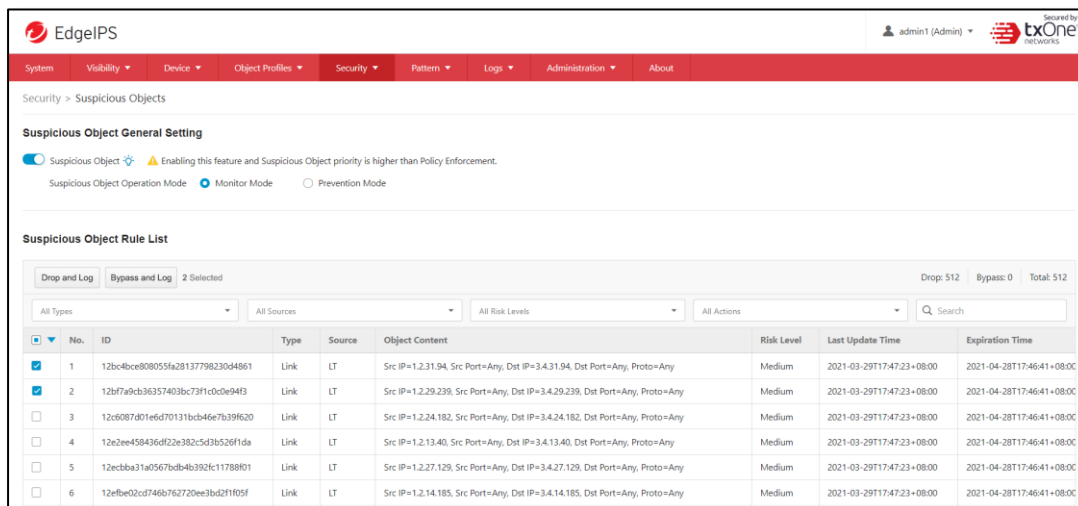
☐ ID	Type	Source	Object Content	Risk Level	Last Update Time	Expiration Time	Action Status
☐ 12123412	Node	DDI-1	10.10.10.1	High	2020-11-22T07:51:49+08:00	2020-11-22T07:51:49+08:00	Drop and Log
☐ 22123412	Link	DDI-2	Src IP=10.100.100.9, Dst IP=192.168.1.1, Proto=6"	Critical	2020-11-22T07:51:49+08:00	2020-11-22T07:51:49+08:00	Bypass and Log
☐ 32123412	Node	Nozomi-1	10.10.10.1	High	2020-11-22T07:51:49+08:00	2020-11-22T07:51:49+08:00	Drop and Log
☐ 42123412	Link	Nozomi-2	Src IP=10.100.100.9, Dst IP=192.168.1.1, Proto=6"	High	2020-11-22T07:51:49+08:00	2020-11-22T07:51:49+08:00	Bypass and Log
☐ 52123412	Link	DDI-2	Src IP=10.100.100.9, Dst IP=192.168.1.1, Proto=6"	Critical	2020-11-22T07:51:49+08:00	2020-11-22T07:51:49+08:00	Drop and Log
☐ 62123412	Node	Nozomi-1	10.10.10.1	Critical	2020-11-22T07:51:49+08:00	2020-11-22T07:51:49+08:00	Drop and Log
☐ 72123412	Link	Nozomi-1	Src IP=10.100.100.9, Dst IP=192.168.1.1, Proto=6"	Critical	2020-11-22T07:51:49+08:00	2020-11-22T07:51:49+08:00	Drop and Log

Note: Before you enable suspicious objects feature, please make sure that the [ODC Setting] pane in [Administration] > [Sync Settings] is properly configured and ODC is connected with 3rd party API from the source of suspicious object.

Configuring Suspicious Objects

Procedure

1. Go to [Security] > [Suspicious Objects].
2. Under the [Suspicious Objects] tab, you will see the [Suspicious Object General Settings] pane.
3. Use the toggle to enable or disable the suspicious object feature.
4. Select a mode ([Monitor Mode], or [Prevention Mode]) for the feature.



Suspicious Object Rule List

Drop and Log | Bypass and Log | 2 Selected

Drop: 512 | Bypass: 0 | Total: 512

No.	ID	Type	Source	Object Content	Risk Level	Last Update Time	Expiration Time
1	12bc4bce808055fa28137798230d44861	Link	LT	Src IP=1.2.31.94, Src Port=Any, Dst IP=3.4.31.94, Dst Port=Any, Proto=Any	Medium	2021-03-29T17:47:23+08:00	2021-04-28T17:46:41+08:00
2	12bf7a9cb36357403bc73f1c0c0e9493	Link	LT	Src IP=1.2.29.239, Src Port=Any, Dst IP=3.4.29.239, Dst Port=Any, Proto=Any	Medium	2021-03-29T17:47:23+08:00	2021-04-28T17:46:41+08:00
3	12c6087d01e6d70131bcb46e7b39f620	Link	LT	Src IP=1.2.24.182, Src Port=Any, Dst IP=3.4.24.182, Dst Port=Any, Proto=Any	Medium	2021-03-29T17:47:23+08:00	2021-04-28T17:46:41+08:00
4	12e2ee458436df22e382c5d3b526f1da	Link	LT	Src IP=1.2.13.40, Src Port=Any, Dst IP=3.4.13.40, Dst Port=Any, Proto=Any	Medium	2021-03-29T17:47:23+08:00	2021-04-28T17:46:41+08:00
5	12ecbba31a0567b0b4b392fc11789f01	Link	LT	Src IP=1.2.27.129, Src Port=Any, Dst IP=3.4.27.129, Dst Port=Any, Proto=Any	Medium	2021-03-29T17:47:23+08:00	2021-04-28T17:46:41+08:00
6	12efbe02cd746b762720ee3bd2f1f05f	Link	LT	Src IP=1.2.14.185, Src Port=Any, Dst IP=3.4.14.185, Dst Port=Any, Proto=Any	Medium	2021-03-29T17:47:23+08:00	2021-04-28T17:46:41+08:00

5. If you want to change the action of a specific suspicious object in the [Suspicious Object Rule List] table, select a specific suspicious object and choose the action [Drop and Log] or [Bypass and Log] when a pattern is matched.

The following table summarizes the settings:

Modes	Suspicious Object Operation Modes	Action Settings	Actions Performed
Inline Mode	Prevention Mode	Prevent and Log	- Blocks network node or network link. - Generates logs.
	Monitor Mode	Monitor and Log	- Detects network node or network link, but does not block communication related to the network node or network link. - Generates logs.
Offline Mode	Prevention Mode / Monitor Mode	Monitor and Log	- Detects network node or network link - Generates logs.

Note: The suspicious objects list takes a higher priority than the device rule and master rule lists.

The Pattern Tab

This chapter describes how to view the pattern information and how to import a TXODI™ (The TXOne One-pass Deep Packet Inspection for Industrial) pattern to the EdgeIPS™ LE device.

The DPI pattern, prepared by TXOne, contains signatures to enable the intrusion prevention features on the device. The intrusion prevention feature detects and prevents behaviors related to network intrusion attempts or targeted attacks at the network level.

Viewing Device Pattern Information

Procedure

1. Go to [Pattern] > [Pattern Update].
2. Under the [Pattern Update] tab, you will see the following pane.



Pattern > Pattern Update

IPS Pattern Update

Pattern Version	TM_IPSP_201209_07
Pattern Build Date	2021-04-08T23:40:22+08:00
Pattern Update File Path	Select Upload
Pattern Information	Release Notes IPS Rule Metadata

3. The [IPS Pattern Update] pane shows the [Pattern Version] and [Pattern Build Date].

Manually Updating the Pattern

Procedure

1. Go to [Pattern] > [Pattern Update].
2. Under the [Pattern Update] tab, you will see the following pane.



Pattern Update File Path Select Upload

3. Click [Select].
4. Manually select the pattern to be deployed to the device.
5. Click [Upload] and then [Confirm].

Downloading Release Info

Procedure

1. Go to [Pattern] > [Pattern Update].
2. Under the [Pattern Update] tab, you will see the following pane.



3. Click the [Release Notes] download button to see the detailed release information.
4. Click the [IPS Rule Metadata] download button to see the detailed IPS rule metadata in a CSV file, including Signature ID, Rule Name, Threat Category, Risk Level, and Default Actions.

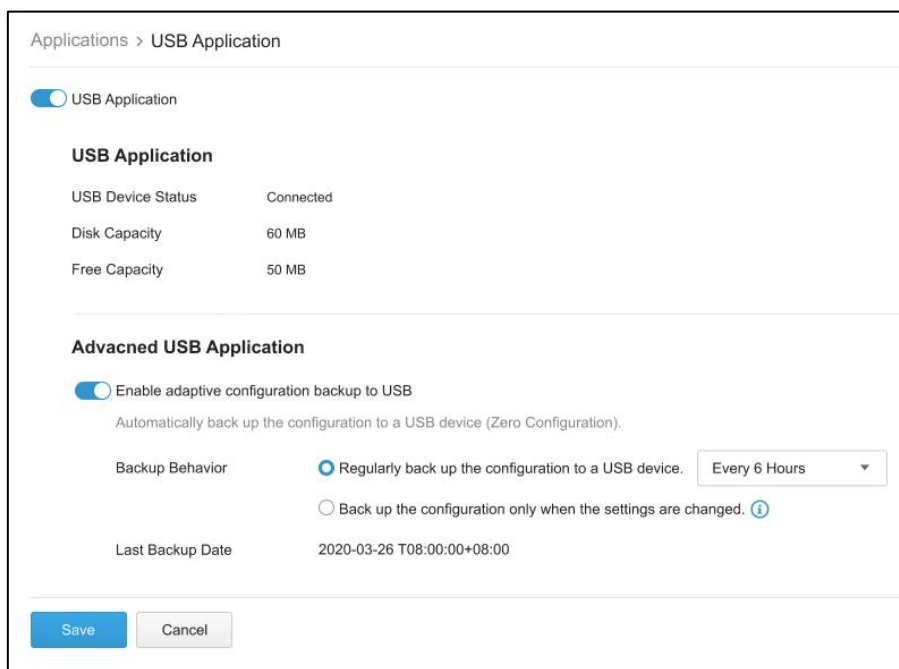
The Application Tab

This chapter describes how to use the USB application and packet capture functions.

USB Application

Procedure

1. Go to [Application] > [USB Application].
2. Under the [USB Application] tab, you will see the following pane.



The screenshot shows the 'Applications > USB Application' configuration pane. At the top, there is a toggle switch for 'USB Application' which is currently turned on. Below this, the 'USB Application' section displays the status of a connected USB device: 'USB Device Status' is 'Connected', 'Disk Capacity' is '60 MB', and 'Free Capacity' is '50 MB'. The 'Advanced USB Application' section contains a toggle for 'Enable adaptive configuration backup to USB', which is also turned on. Below this toggle is a description: 'Automatically back up the configuration to a USB device (Zero Configuration)'. The 'Backup Behavior' section has two radio button options: 'Regularly back up the configuration to a USB device.' (selected) and 'Back up the configuration only when the settings are changed.' (with an information icon). A dropdown menu next to the selected option shows 'Every 6 Hours'. The 'Last Backup Date' is displayed as '2020-03-26 T08:00:00+08:00'. At the bottom of the pane are 'Save' and 'Cancel' buttons.

3. Click [Enable] to enable USB Application usage. This toggle switch controls whether the USB port is enabled or not.
4. Once USB Application is enabled and a USB disk is plugged in, you can see the status and information about disk capacity and remaining free space.

Note: If USB Application is disabled, the USB port on the front panel will not connect to the plugged-in USB devices.

Note: Regarding supported USB devices, please refer to [Supported USB Devices on page 63](#).

Advanced USB Application

1. Click [Enable] to enable adaptive configuration backup for a USB-based device.
2. Back up behavior can be configured as follows:
 - a. Periodic backup of a configuration to USB – 6 different time periods are supported.
 - b. Backup configuration to a USB disk when configurations are changed.

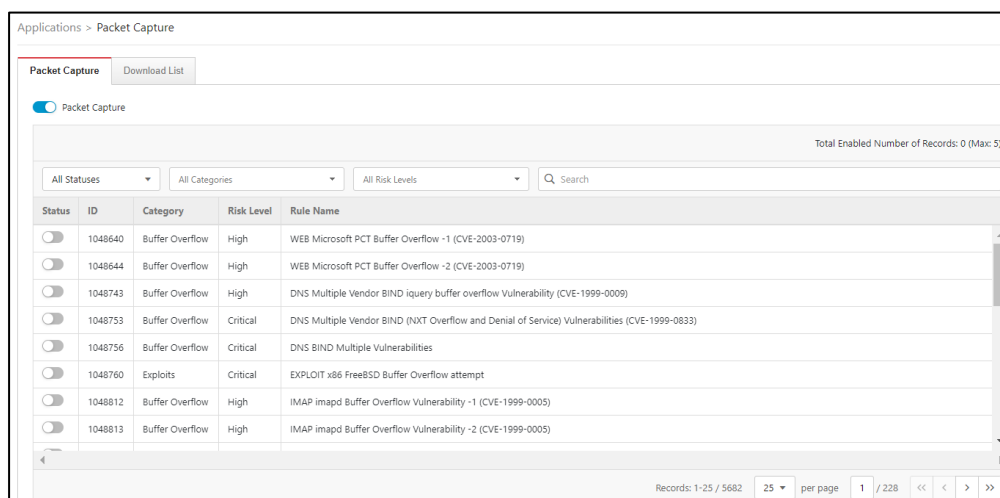
Packet Capture

The Packet Capture feature allows you to capture packets for further analysis and to configure the capture of packets by IPS event rules. Packets that trigger IPS events can then be further analyzed and can help support teams to quickly address false positive/false negative matching of IPS rules in the security module.

Enabling Packet Capture

Procedure

1. Go to [Application] > [Packet Capture].
2. Click [Enable] to enable IPS packet capture.



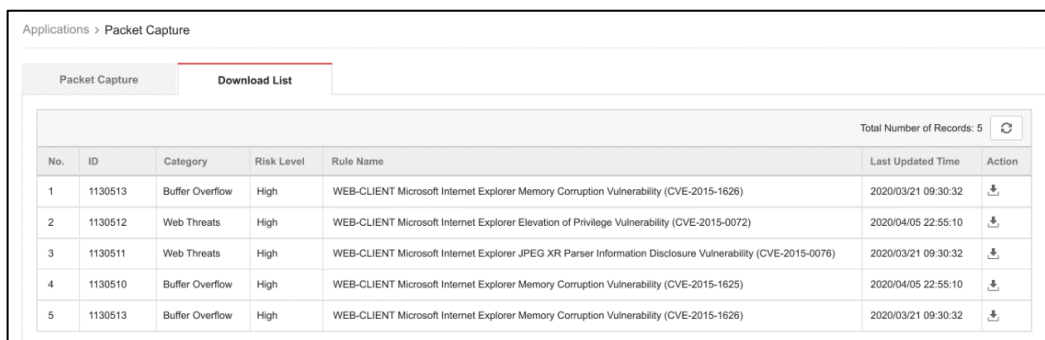
3. You can see the entire IPS rule list and select a rule to "Enable" for IPS rule capture.
4. Up to 10 rules can be selected for IPS Rule packet capture support.

Note: The packet capture feature will save the selected IPS Rule event packets once the IPS rule is hit and will only save the last 5 occurrences of a particular rule. Older events will be overwritten.

Download Captured Packet

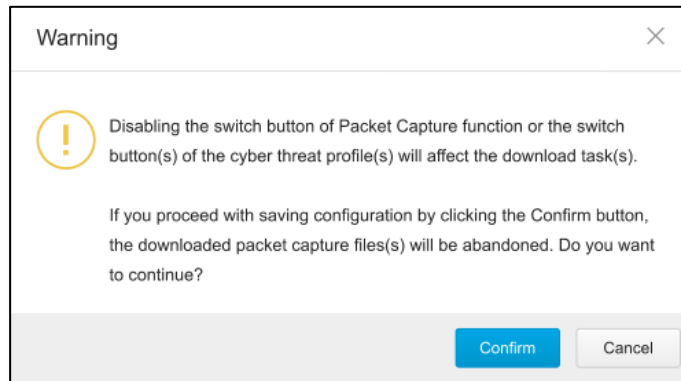
Procedure

1. Go to [Application] > [Packet Capture].
2. Click [Download List] to show a list of IPS rules where you can download a zip archive of



each rule's related pcap files.

3. You can click the download icon to download the zipped archive to your disk.
4. Disabling packet capture will cause previously downloaded packet captures to be deleted. To confirm disabling of the feature, the above warning will be shown to the user.



Note: The download list will be refreshed every 10 seconds. If you want to get the latest updates, please click the "manual" refresh button.

The Logs Tab

This chapter describes the system event logs and security detection logs you can view on the management console.

You can view the following logs on the operational technology defense console:

- **Cyber Security Logs**
- **Policy Enforcement Logs**
- **Protocol Filter Logs**
- **File Filter Logs**
- **Suspicious Object Logs**
- **Assets Detection Logs**
- **System Logs**
- **Audit logs (Can only be viewed by auditor)**

Viewing Cyber Security Logs

"Cyber Security Logs" will include logs detected by both intrusion prevention and denial of service prevention features.

Procedure

1. Go to [Logs] > [Cyber Security Logs].

The following table describes the cyber security log information.

Fields	Descriptions
Time	The time when the log entry was created.
Rule Name	The name of the policy enforcement rule that was used to generate the log.
Profile Name	The name of the IPS profile that was used to generate the log.
Event ID	The ID of the matched signature.
TID	The identifier for an item in the Mitre ATT&CK knowledge base.
Security Category	The category of the matched signature.
Security Severity	The severity level assigned to the matched signature.
Security Rule Name	The name of the matched signature.
Interface	The physical port interface which receives the cyber attack
Attacker	The IP address of host device which initiates the cyber attack
Source MAC Address	The source MAC address of the connection.
Source IP Address	The source IP address of the connection.
Source Port	The source port of the packet, if protocol is TCP/UDP. The ICMP type of the packet, if protocol is ICMP.
Destination MAC address	The destination MAC address of the connection.
Destination IP Address	The destination IP address of the connection.
Destination Port	The destination port of the packet, if protocol is TCP/UDP. The ICMP type of the packet, if protocol is ICMP.
VLAN ID	The VLAN ID of the connection.
Ethernet Type	The Ethernet type of the connection.

Fields	Descriptions
IP Protocol Name	The IP protocol name of the connection.
Action	The action performed based on the IPS profile settings.
Count	The number of detected network packets within the detection period after the detection threshold is reached.

Viewing Policy Enforcement Logs

The policy enforcement logs cover logs created by the [Policy Enforcement] feature without [Protocol Filter] being enabled, i.e., the [Action] of the policy enforcement rule is either to allow or to deny. The protocol filter is not used in the policy rule.

Procedure

1. Go to [Logs] > [Policy Enforcement Logs].

The following table describes the policy enforcement log information.

Fields	Descriptions
Time	The time when the log entry was created.
Rule Name	The name of the policy enforcement rule set and the matched policy rule that was used to generate the log.
Interface	The physical port interface which receives the traffic and the traffic matches the policy enforcement rule.
Source MAC Address	The source MAC address of the connection.
Source IP Address	The source IP address of the connection.
Source Port	The source port, if protocol is TCP/UDP. The ICMP type, if protocol is ICMP.
Destination MAC Address	The destination MAC address of the connection.
Destination IP Address	The destination IP address of the connection.
Destination Port	The destination port, if protocol is TCP/UDP. The ICMP code, if protocol is ICMP.
VLAN ID	The VLAN ID of the connection.
IP Protocol Name	The IP protocol name of the connection.
Action	The action performed based on the policy settings.

Viewing Protocol Filter Logs

The protocol filter logs cover logs detected by the [Protocol Filter] feature. Protocol filter is an advanced setting that can be configured under the [Policy Enforcement] settings.

Procedure

1. Go to [Logs] > [Protocol Filter Logs].

The following table describes the protocol filter log information.

Fields	Descriptions
Time	The time when the log entry was created.
Rule Name	The name of the policy enforcement rule that was used to generate the log.
Profile Name	The name of the protocol filter profile that was used to generate the log.
Interface	The physical port interface which receives the traffic and the traffic matches the setting of protocol filter profile.
Source MAC Address	The source MAC address of the connection.

Fields	Descriptions
Source IP Address	The source IP address of the connection.
Source Port	The source port, if protocol is TCP/UDP. The ICMP type, if protocol is ICMP.
Destination MAC address	The destination MAC address of the connection.
Destination IP Address	The destination IP address of the connection.
Destination Port	The destination port, if protocol is TCP/UDP. The ICMP code, if protocol is ICMP.
VLAN ID	The VLAN ID of the connection.
Ethernet Type	The Ethernet type of the connection.
IP Protocol Name	The IP protocol name of the connection.
L7 Protocol Name	The layer 7 protocol name of the connection. The term layer 7 refers to the one defined in the OSI (Open Systems Interconnection) model.
Cmd / Fun No.	The command or function number that triggered the log.
Extra Information	Extra information provided with the log.
Action	The action performed based on the policy settings.
Count	The number of detected network packets.

Viewing File Filter Logs

"File Filter Logs" refers to logs detected by the [File Filter] feature. The file filter is an advanced setting that can be configured under [Policy Enforcement] settings.

Procedure

1. Go to [Logs] > [File Filter Logs]

The following table describes the file filter log information.

Fields	Descriptions
Time	The time when the log entry was created.
Rule Name	The name of the policy enforcement rule set and the matched policy rule that was used to generate the log.
Profile Name	The name of the file filter profile that was used to generate the log.
Interface	The physical port interface which receives the packet.
Source MAC Address	The source MAC address of the packet.
Source IP Address	The source IP address of the packet.
Source Port	The source port, if the protocol is TCP/UDP.
Destination MAC address	The destination MAC address of the packet.
Destination IP Address	The destination IP address of the packet.
Destination Port	The destination port, if protocol is TCP/UDP.
VLAN ID	The VLAN ID of the connection.
Protocol	The layer 7 protocol name of the connection. The term layer 7 refers to the one defined in the OSI (Open Systems Interconnection) model.
Extra Information	Extra information provided with the file filter log.
Action	The action performed based on the policy settings.

Viewing Suspicious Object Logs

The suspicious object logs include logs detected by the [Suspicious Objects] feature.

Procedure

1. Go to [Logs] > [Suspicious Object Logs].

The following table describes the suspicious object log information.

Fields	Descriptions
Time	The time when the log entry was created.
ID	The hash ID of the matched suspicious object.
Type	The suspicious object type, which is node type or link type.
Source	The source of the suspicious object
Risk Level	The threat level of the suspicious object
Expiration Time	The expiration time of the suspicious object. When the expiration time reaches, the suspicious object will be deleted.
Interface	The physical port interface which receives the suspicious object
Source MAC Address	The source MAC address of the connection.
Source IP Address	The source IP address of the connection.
Source Port	The source port of the connection.
Destination MAC address	The destination MAC address of the connection.
Destination IP Address	The destination IP address of the connection.
Destination Port	The destination port of the connection.
VLAN ID	The VLAN ID of the connection.
Ethernet Type	The Ethernet type of the connection.
IP Protocol Name	The IP protocol name of the connection.
Action	The action performed based on the suspicious object settings.
Count	The number of detected suspicious object within the detection threshold.

Viewing Asset Detection Logs

The asset detection logs cover the system status changes of the managed assets.

Procedure

1. Go to [Logs] > [Asset Detection Logs].

The following table describes the asset detection log information.

Fields	Descriptions
Time	The time when the log entry was created.
Event Type	The log event description.
Interface	The physical port interface which receives the asset information.
Asset MAC Address	The MAC address of the asset.
Asset IP Address	The IP address of the asset.

Viewing System Logs

You can view details about system events on the device.

Procedure

1. Go to [Logs] > [System Logs].

The following table describes the system log information.

Fields	Descriptions
Time	The time when the log entry was created.
Severity	The severity level of the logs.
Message	The log event description.

Viewing Audit Logs

You can view details about user access, configuration changes, and other events that occurred when using the device.

Procedure

1. Go to [Logs] > [Audit Logs].

The following table describes the audit log information.

Fields	Descriptions
Time	The time when the log entry was created.
User ID	The user account used to execute the task.
Client IP	The IP address of the host used to access the management console.
Severity	The severity level of the logs.
Message	The log event description.

Note: To view audit logs, please log in with the default "audit" account.

The Administration Tab

This chapter describes the available administrative settings for EdgeIPS™ LE device.

Account Management

Note: Log on to the management console with an administrator account to access the Accounts tab.

This system uses role-based administration to grant and control access to the management console. Use this feature to assign specific management console privileges to user accounts and present them with only the tools and permissions necessary to perform specific tasks. Each account is assigned a specific role. A role defines the access level on the management console. Users log on to the management console by using custom user accounts.

The following table outline the tasks available under the [Account Management] tab.

Tasks	Descriptions
Add an account	Click Add to create a new user account. For more information, see Adding a User Account on page 48 .
Delete an existing account	Select an existing user account and click Delete.
Edit an existing account	Click the name of an existing user account to view or modify the current account settings.

User Roles

The following table describes the permissions matrix for user roles.

Sub-Tabs	Actions	User Roles			
		Admin	Operator	Viewer	Auditor
System	View	Yes	Yes	Yes	Yes
	All operations	Yes	Yes	Yes	Yes
Visibility	View	Yes	Yes	Yes	No
	All operations	Yes	Yes	Yes	No
Device	View	Yes	Yes	No	No
	All operations	Yes	No	No	No
Object Profiles	View	Yes	Yes	No	No
	All operations	Yes	Yes	No	No
Security	View	Yes	Yes	No	No
	All operations	Yes	Yes	No	No
Pattern	View	Yes	Yes	No	No
	All operations	Yes	Yes	No	No
Logs – excluding Audit Logs	View	Yes	Yes	Yes	No
Audit Log	View	No	No	No	Yes
Administration	View	Yes	No	No	No
	All operations	Yes	No	No	No

Built-in User Accounts

The following table lists the built-in user accounts in the device.

Built-in default Account IDs	User Roles	Default Passwords
admin	Admin	txone
auditor	Auditor	txone

Note: The built-in user accounts cannot be deleted from the device. Ensure that the passwords of the built-in accounts are changed when you first set up the device.

Adding a User Account

When you log on using the administrator account, you can create new user accounts to access the system.

Procedure

1. Go to [Administration] > [Account Management].
2. Click [Add].
The Add User Account screen will appear.
3. Configure the account settings.

Fields	Descriptions
ID	Type the user ID to log on to the management console.
Name	Type the name of the user for this account.
Authentication Source	Type the authentication source for this account
Local Password	Type the account password stored in the device.
Confirm password	Type the account password again to confirm.
Description	Add a description for this account
Role	Select a user role for this account. For more information, see User Roles on page 47 .

4. Click [Save].

Changing Your Password

Procedure

1. On the management console banner, click your account name.
2. Click [Change Password].
The Change Password screen will appear.
3. Configure the password settings.
 - Current password
 - New password
 - Re-type password for confirmation
4. Click [OK].

Configuring Password Policy Settings

EdgeIPS™ LE provides the following password policy settings to enhance web console access security:

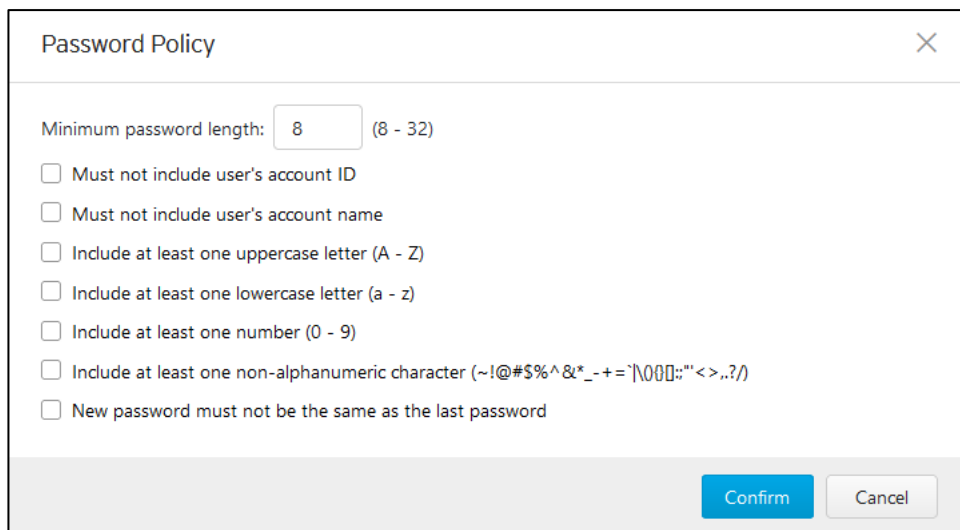
- **Password complexity settings**

Specify password complexity settings to enforce strong passwords. For example, you can specify that users must create strong passwords containing a combination of both uppercase and lowercase letters, numbers, and symbols, and the passwords must be at least eight characters in length.

Note: If strong passwords are required, when users submit new passwords, the password policy determines whether the passwords meet your company's established requirements. Strict password policies may sometimes increase costs to an organization when users select passwords that are too difficult to remember. Users would call the help desk when they forget their passwords, or keep passwords in easily accessible locations that would increase their vulnerability to threats. When establishing a password policy, balance your need for strong security against the need to make the policy easy for users to follow.

Procedure

1. Go to [Administration] > [Account Management].
2. Click the [Password Policy] tab.
The [Password Policy] screen will appear.



3. Select one or multiple options that meet your required password policy.
4. Click Save.

Auth Service

Use the [Auth Service] tab to do the following:

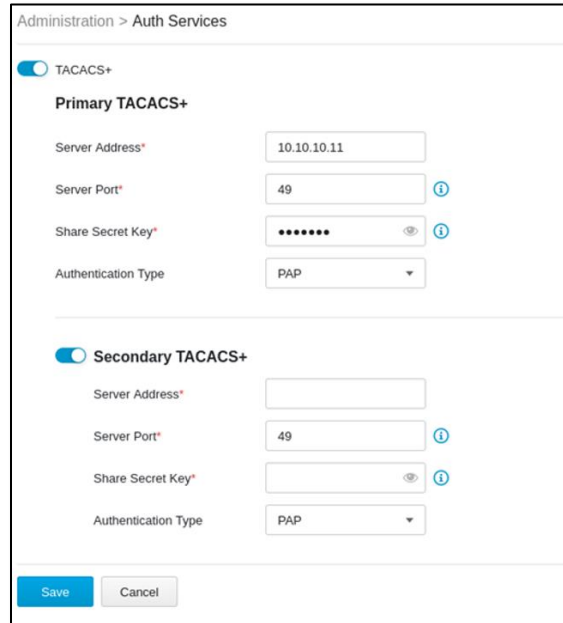
- Configure the TACACS+ of the device.

Configuring TACACS+

Procedure

1. Go to [Administration] > [Auth Services].

2. In the [TACACS] pane, provide the Primary and Secondary TACACS+ Server for the device.



Administration > Auth Services

☒ TACACS+

Primary TACACS+

Server Address* 10.10.10.11

Server Port* 49 ⓘ

Share Secret Key* ⓘ

Authentication Type PAP ▼

Secondary TACACS+

Server Address*

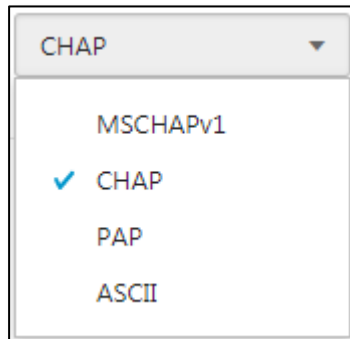
Server Port* 49 ⓘ

Share Secret Key* ⓘ

Authentication Type PAP ▼

Save Cancel

3. Enable Primary TACACS+ and configure settings
 - a. Configure Server address.
 - b. Configure Server Port (Default port: 49)
 - c. Configure Share Secret Key (MAX: 64 characters)
 - d. Select Authentication type, includes list as follows



CHAP ▼

MSCHAPv1

✓ CHAP

PAP

ASCII

- e. Enable Secondary TACACS+ Server if necessary

The System Management Tab

Use the [System Management] tab to do the following:

- Configure the host name and location information of the device.
- Configure the IP addresses that are allowed to manage the device.
- Choose the protocols and ports that can be used to manage the device.

Configuring Device Name and Device Location Information

Procedure

1. Go to [Administration] > [System Management].
2. In the [System Settings] pane, provide the host name and location information of the device.



System Settings

Hostname* ⓘ

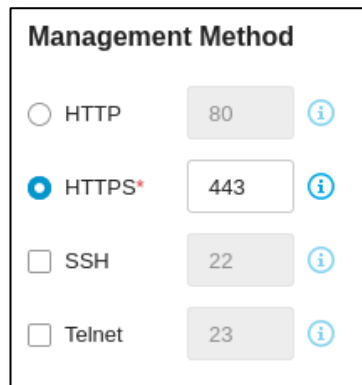
Location Information ⓘ
 (Sample: Zone1, Network-1)

Configuring Management Method and Access Control List

Configuring Management Protocols and Ports

Procedure

1. Go to [Administration] > [System Management].
2. In the [Management Method] pane:
 - a. Select the protocols that are allowed to be used.
 - b. Input the port numbers for the protocols.



Management Method

☐ HTTP ⓘ

☒ HTTPS* ⓘ

☐ SSH ⓘ

☐ Telnet ⓘ

Note: The HTTP and HTTPS protocols are used for connecting to the web management console. The SSH and Telnet protocols are used for connecting to the CLI commands.

Configuring Control List Access from Management Clients

Procedure

1. Go to [Administration] > [System Management].
2. In the [Management Method] pane, use the toggle to enable or disable access control from the management clients.
3. List the IP addresses that are allowed to manage the device.

☒ Access Control List

Allowed IP 1

Allowed IP 2

Allowed IP 3

Allowed IP 4

4. Check if pings to the Management Interface are allowed.

The Sync Settings Tab

EdgeIPS[™] LE can be managed by ODC (Operational Technology Defense Console). Use this tab to register the EdgeIPS[™] LE to ODC.

Enabling Management by ODC

Procedure

1. Go to [Administration] > [Sync Settings].
2. In the pane:
 - a. Use the toggle to enable management by ODC.
 - b. Input the IP address of the ODC server.

Administration > Sync Settings

☒ Enable ODC Management

ODC Server Address

ODC Sync Connected

Action for Master Rule and Profile

Procedure

1. Go to [Administration] > [Sync Settings].
2. Click the [Master Rule & Profile] button to delete Object Profiles and Policy Enforcement rules synced from ODC

Action for Master Rule and Profile

Purge the imported Master Rule(s) and Profile(s) when they are not suitable for re-deployment.

The Syslog Tab

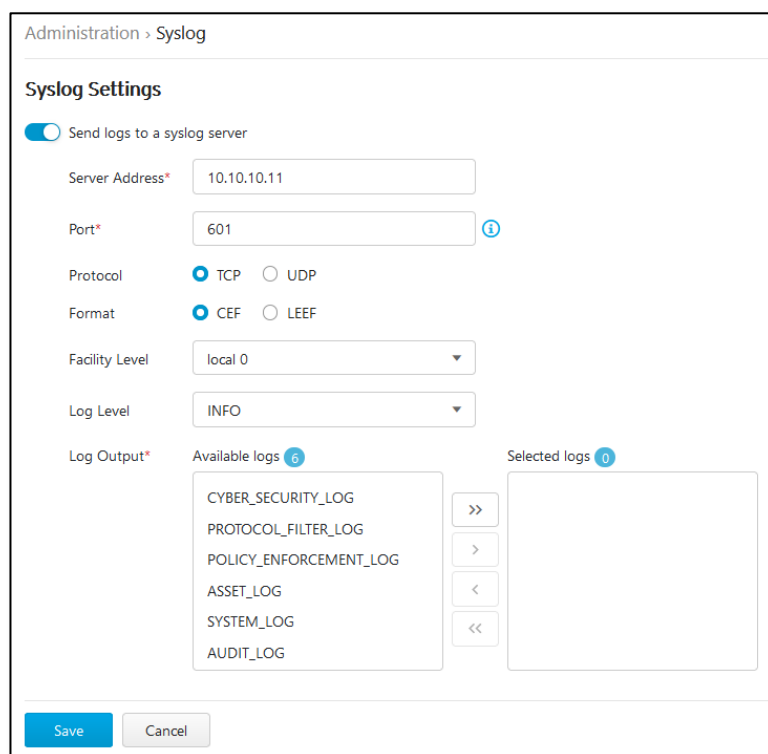
The EdgeIPS™ LE system maintains Syslog events that provide summaries of security and system events. Common Event Format (CEF) and Log Event Extended Format (LEEF) syslog messages are used in EdgeIPS™ LE.

Configure the Syslog settings to enable the device to send the Syslog to a Syslog server.

Configuring Syslog Settings

Procedure

1. Go to [Administration] > [Syslog].



2. Select [Send logs to a syslog server] to set the EdgeIPS™ LE to send logs to a Syslog server.
3. Configure the following settings.

Fields	Descriptions
Server address	Type the IP address of a Syslog server.
Port	Type a port number.
Protocol	Select a protocol for the communication.
Format	Select a syslog format: CEF or LEEF
Facility Level	Select a facility level to determine the source and priority of the logs.
Log Level	Select a Syslog severity level. This device only sends logs with the selected severity level or higher to the Syslog servers. For more information, see Syslog Severity Levels on page 54 .

4. Select the types of logs to send.
5. Click Save.

Syslog Severity Levels

The Syslog severity level specifies the type of messages to be sent to the Syslog server.

Levels	Severities	Descriptions
0	Emergency	Complete system failure Take immediate action.
1	Alert	Primary system failure Take immediate action.
2	Critical	Urgent failure Take immediate action.
3	Error	Non-urgent failure Resolve issues quickly.
4	Warning	Error pending Take action to avoid errors.
5	Notice	Unusual events Immediate action is not required.
6	Informational	Normal operational messages useful for reporting, measuring throughput, and other purposes No action is required.
7	Debug	Useful information when debugging the application. Note: Setting the debug level can generate a large amount of Syslog traffic in a busy network. Use with caution.

Syslog Severity Level Mapping Table

The following table summarizes the logs of Policy Enforcement/Protocol Filter/Cyber Security and their equivalent Syslog severity levels.

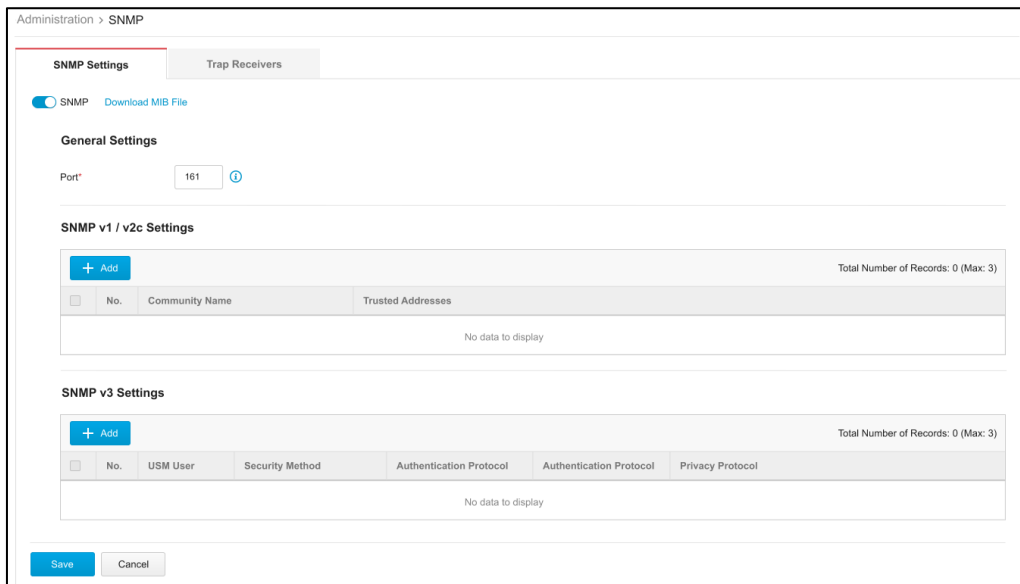
Policy Enforcement / Protocol Filter Actions	Cyber Security Severity Levels	Syslog Severity Levels
		0 - Emergency
	Critical	1 - Alert
	High	2 - Critical
		3 - Error
Deny	Medium	4 - Warning
		5 - Notice
Allow		6 - Information
		7 - Debug

The SNMP Tab

The Simple Network Management Protocol is a protocol used for exchanging management information among Edge series devices. EdgeIPS™ LE supports SNMP v1/v2c and a more secure v3, as well as SNMP traps.

Procedure

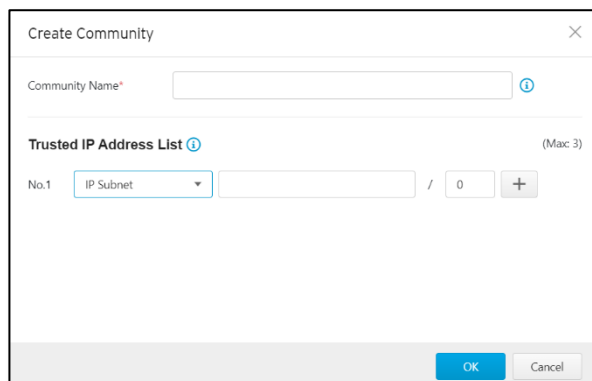
1. Go to [Administration] > [SNMP]
2. Click [Enable] to enable SNMP function.
3. Under General settings, you can change the SNMP port. The default setting is Port 161.
4. You can click the "Download MIB file" link to download the EdgeIPS™ LE's MIB file.



Configuring SNMP v1/v2c

Procedure

1. Go to [Administration] > [SNMP]
2. Click [Add] to create SNMP v1/v2c settings.

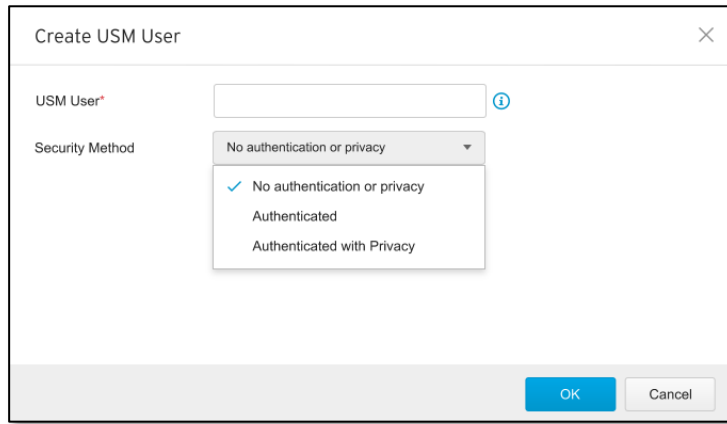


- a. Enter Community name
- b. Add a Trusted Address list. There are two supported types: Single IP and IP Subnet
- c. Click [OK] to create a new SNMP v1/v2c community

Configuring SNMP v3

Procedure

1. Go to [Administration] > [SNMP]
2. Click [Add] to create SNMP v3 settings.



The 'Create USM User' dialog box contains the following fields and options:

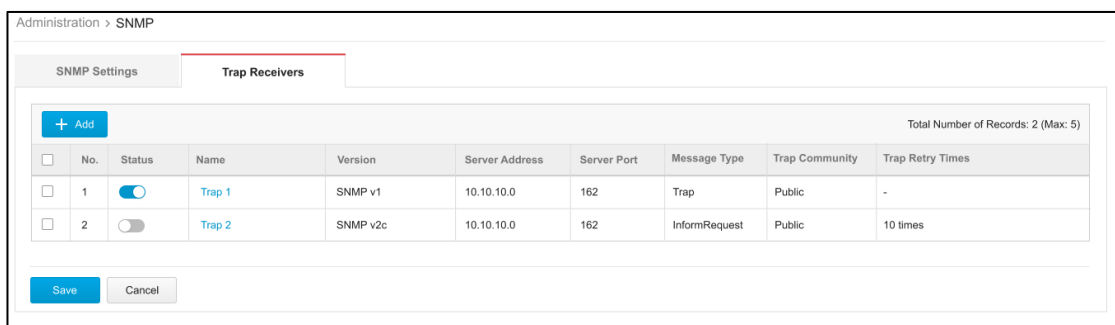
- USM User***: A text input field with an information icon.
- Security Method**: A dropdown menu with the following options:
 - No authentication or privacy (selected with a checkmark)
 - Authenticated
 - Authenticated with Privacy
- Buttons**: 'OK' and 'Cancel' buttons at the bottom right.

3. Enter USM user.
4. Under [Security Method] drop-down list, select one of the following options:
 - a. No authentication or privacy.
 - b. Authenticated – this includes SHA and MD5. You can select an appropriate authentication protocol and enter an Authentication Key.
 - c. Authenticated with Privacy – this also includes SHA and MD5, and you can select appropriate authentication and privacy protocols.
5. Click [OK] to create an SNMPv3 USM User.

Configuring SNMP Trap Receivers

Procedure

1. Go to [Administration] > [SNMP]
2. Click the [Trap Receivers] tab.



The 'Trap Receivers' configuration interface shows a table with the following data:

No.	Status	Name	Version	Server Address	Server Port	Message Type	Trap Community	Trap Retry Times
1	☒	Trap 1	SNMP v1	10.10.10.0	162	Trap	Public	-
2	☐	Trap 2	SNMP v2c	10.10.10.0	162	InformRequest	Public	10 times

Buttons: '+ Add', 'Save', 'Cancel'. Total Number of Records: 2 (Max: 5)

3. Click [Add] to create a new Trap Receiver.
 - a. Click the toggle under [Status] to enable a Trap Receiver.
 - b. Enter [Name] to create a Trap Receiver name.
 - c. Add [Description] if necessary.
 - d. Select SNMP version – options include SNMP v1 and SNMP v2c
 - e. Enter [Server Address]

- f. Enter [Server Port]. The default setting is port 162.
- g. Select message type, "Trap" and "informRequest".
- h. Enter Trap Community, default name: PUBLIC.
- i. Set the Trap Retry Times to an amount of retries ranging from 1 to 10 times.

Create Trap Receiver
✕

Status ☒

Name* ?

Description ?

Version ☐ SNMP v1 ☒ SNMP v2c

Server Address*

Server Port* ?

Message Type ☒ Trap ☐ InformRequest

Trap Community*

Trap Retry Times

- j. Select what will trigger an Event Notification from the list as follows:

Event Notification*

- ☐ High CPU Usage ?
- ☐ High Memory Usage
- ☐ Low Disk Space for Logs
- ☐ Interface IP Address Changed
- ☐ Network Interface Linked Up
- ☐ Network Interface Linked Down

Note: When the CPU usage reaches 70%, 80% or 95%, the system will send an event notification.

Note: When the memory usage reaches 80%, the system will send an event notification.

Note: When the log storage reaches 95%, the system will keep sending event notifications until the log storage is below 95%.

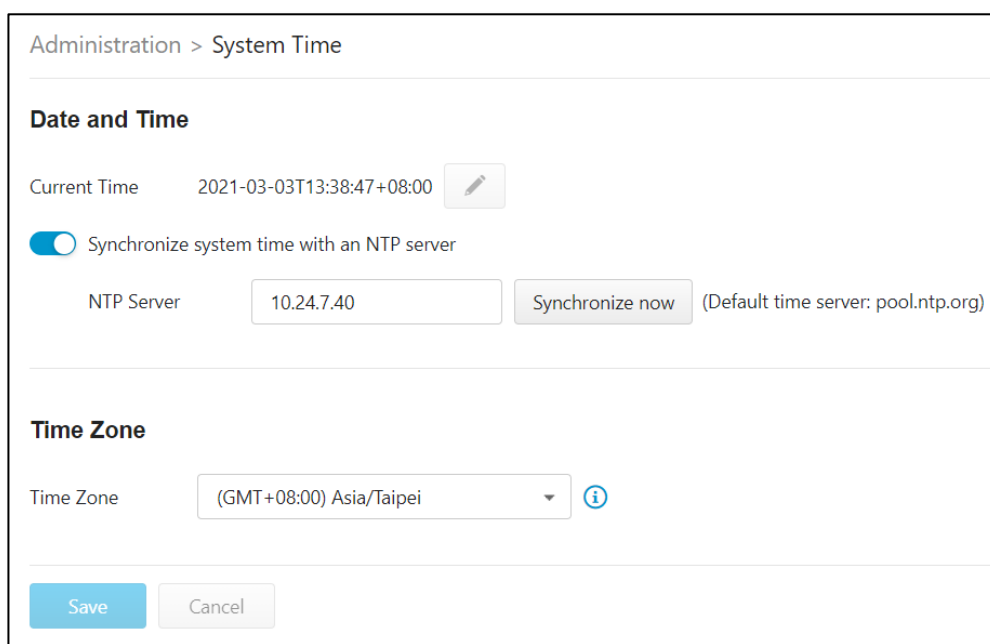
The System Time Tab

Network Time Protocol (NTP) synchronizes computer system clocks across the Internet. Configure NTP settings to synchronize the server clock with an NTP server, or manually set the system time.

Configuring System Time

Procedure

1. Go to [Administration] > [System Time].



2. In the [Date and Time] pane, select one of the following methods:
 - Synchronize system time with an NTP server
 - a. Specify the domain name or IP address of the NTP server.
 - b. Click Synchronize Now.
 - Set system time manually
 - a. Click the calendar to elect the date and time.
 - b. Set the hour, minute, and second.
 - c. Click Apply.
3. From the [Time Zone] drop-down list, select a time zone.
4. Click Save.

Note: ODC system synchronizes the system time with its managed instances.

The Back Up / Restore Tab

Export settings from the management console to back up the configuration of your EdgeIPS™ LE. If a system failure occurs, you can restore the settings by importing the configuration file that you previously backed up.

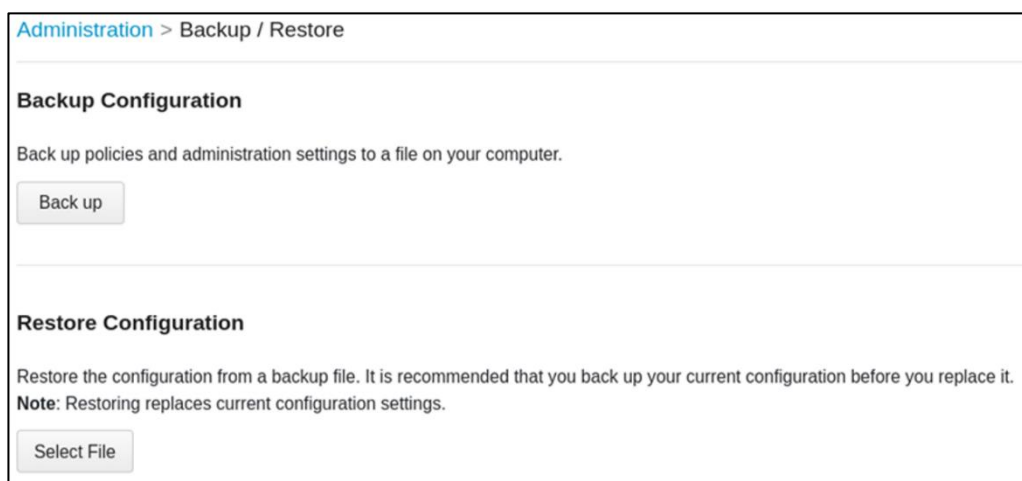
We recommend the following:

- Backing up the current configuration before each import operation.
- Performing the operation when the EdgeIPS™ LE is idle. Importing and exporting configuration settings affect the performance of EdgeIPS™ LE.

Backing Up a Configuration

Procedure

1. Go to [Administration] > [Backup / Restore].
The [Backup / Restore] tab will appear.



The screenshot shows a web interface for the 'Backup / Restore' tab. At the top, a breadcrumb trail reads 'Administration > Backup / Restore'. Below this, the 'Backup Configuration' section contains the instruction 'Back up policies and administration settings to a file on your computer.' and a 'Back up' button. The 'Restore Configuration' section contains the instruction 'Restore the configuration from a backup file. It is recommended that you back up your current configuration before you replace it.' followed by a 'Note: Restoring replaces current configuration settings.' and a 'Select File' button.

2. Click the [Back up] button.
A configuration backup file will automatically be saved in your computer.

Restoring a Configuration

Follow the steps below to restore the configurations of the EdgeIPS™ LE.

Procedure

1. Go to [Administration] > [Backup / Restore].
2. In the [Restore Configuration] section, click the [Select File] button, and proceed to import the file.

All services will restart. It may take some time to restart services after applying imported settings and rules.

The Firmware Management Tab

Use the [Firmware Management] tab to:

- View the firmware information for the device
- Upgrade the firmware of the device
- Boot into standby partition and firmware

Viewing Device Firmware Information

Procedure

1. Go to [Administration] > [Firmware Management].
2. The [Firmware Management] pane lists the two available partitions. It shows the [Partition #], [Partition Name], [Partition Status], [Firmware Version] and [Firmware Build Time].

Note: EdgeIPS™ LE can have up to two versions of firmware installed. Each firmware is installed in its own separate partition. At any given point in time, one partition will have the status [Running], which indicates the currently running and active firmware. The other partition will have the status [Standby], which indicates an alternative or standby partition.

Administration > Firmware Management

No.	Partition Name	Partition Status	Firmware Version	Firmware Build Time	Actions
1	boot1	Running	IPSLE_T01_1.0.5	2021-03-25T19:11:33+08:00	
2	boot2	Standby	IPSLE_T01_1.0.4	2021-03-24T17:24:22+08:00	

Upgrading Firmware

Procedure

1. Go to [Administration] > [Firmware Management].

Note: During a firmware upgrade, the firmware will always be installed to the [Standby] partition. As such, the firmware upgrade button is only available in the [Standby] partition row.

Administration > Firmware Management

No.	Partition Name	Partition Status	Firmware Version	Firmware Build Time	Actions
1	boot1	Running	IPSLE_T01_1.0.5	2021-03-25T19:11:33+08:00	
2	boot2	Standby	IPSLE_T01_1.0.4	2021-03-24T17:24:22+08:00	

2. Click on the Upgrade Firmware button to install it to the [Standby] partition.
3. In the [Upgrade Firmware] pane provide the location of the firmware and click [Upload] to install the firmware to the [Standby] partition].

Upgrade Firmware

Firmware Information

Current Firmware Version IPSLE_T01_1.0.4
Firmware Build Time 2021-03-24T17:24:22+08:00

Firmware Update

Local Firmware Update

- After successfully installing the required firmware to [Standby] partition, click on the [Reboot and Apply Firmware] button as shown in the next section.

Note: Various versions of the firmware can be downloaded at the Trend Micro Download Center at https://www.trendmicro.com/en_us/business/products/downloads.html

Rebooting and Applying Firmware

To boot into an upgraded firmware or to revert to the previous firmware, you would need to boot into the [Standby] partition and load the firmware from there.

Procedure

- Go to [Administration] > [Firmware Management].
- Click on the [Reboot and Apply Firmware] button in the [Standby] partition row.

No.	Partition Name	Partition Status	Firmware Version	Firmware Build Time	Actions
1	boot1	Running	IPSLE_T01_1.0.5	2021-03-25T19:11:33+08:00	
2	boot2	Standby	IPSLE_T01_1.0.4	2021-03-24T17:24:22+08:00	

Warning



The standby firmware will become the running firmware after system reboot. Do you want to reboot the system?

Note: Please backup current system configuration before switching firmware partition.

- Click [OK] to proceed with rebooting into the [Standby] partition to make it the [Running] partition.

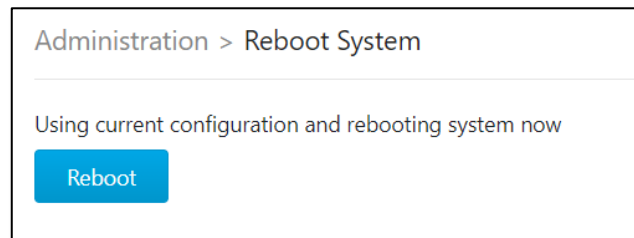
The Reboot System Tab

Use the [Reboot System] tab to reboot the system.

Rebooting the System

Procedure

1. Go to [Administration] > [Reboot System].
2. In the [Reboot System] pane, click [Reboot] to reboot the system.



Supported USB Devices

This chapter describes the use of supported USB devices with the EdgeIPS™ LE device for extended or supporting functionality.

To ensure optimal operation, only the USB devices below are currently supported. The following list may be updated from time to time. Please visit TrendMicro's support page for an up-to-date list.

#	Models		Device Types
1	MOXA Backup Configurator (ABC-02 Series) Model: ABC-02-USB-T		USB Disk Drive
2	Innodisk Industrial USB 2.0 16GB (USB Drive 2SE)		USB Disk Drive
3	Apacer industrial USB disk 16GB (AH355)		USB Disk Drive

Supported actions via USB Disk

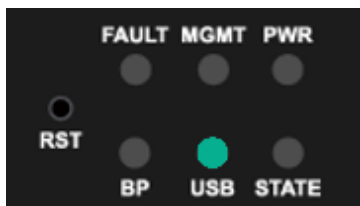
1. On-demand configuration for disk backup
2. Load pattern from disk
3. Load configuration from disk
4. Load firmware from disk

Note: Given that this feature allows anyone with a supported USB disk device to perform various operation via USB, the physical security of the EdgeIPS™ LE device must be considered carefully.

Note: Only supported USB disk devices may be used for this feature.

To perform any of these actions:

1. Plug the supported USB disk device into the EdgeIPS™ LE device's USB port.
2. Upon successful detection of the USB disk device, the "USB" LED indicator will change to a steady green. The system log can also be checked to confirm that a supported USB disk device was detected when inserted. This state is referred to as the "Default Action" state.



Note: If an unsupported USB device is plugged in, it will simply be ignored and no further action will be taken.

The function of the reset button will also change until the USB device is unplugged. When a USB device is plugged in, the reset button will not serve as the reboot/factory reset button. It will instead serve as a button to cycle through a set of possible actions that may be taken during this time.

Users can use the reset button to cycle through a set of possible actions. The LEDs will indicate which action is currently selected. Each quick press of the Reset button will toggle through the next possible action.

Possible Actions to Toggle Through

States/Actions	LEDs	COLOURS/STATES
Default State – USB Plugged in Backup Configuration	USB LED	Green – Steady
Load/Restore Pattern	STATE LED	Green – Blinking (1/sec)
Load/Restore Configuration	MGMT LED	Green – Blinking (1/sec)
Load/Restore Firmware	STATE LED & MGMT LED	Green – Blinking (1/sec)

1. After selecting an action, you must confirm the action by pressing the Reset button for more than 3 seconds (a long and steady press).

Note: The action must be confirmed within 10 seconds. Otherwise, the LEDs will be return to their default state.

2. When an action is being attempted, if there is a USB disk data transfer, the following LEDs will indicate it as shown below and then return to previous state after data transfer is complete.

Data Transfer Indication	LED	COLOUR/STATE
	USB LED	Green – Blinking (Once every 0.5 sec)

3. If any error occurs when an action is being attempted, the following LEDs will show it like so:

Error Indication (on any error while action was being processed)	LED	COLOUR/STATE
	FAULT LED	RED – Steady

Note: The error can only be cleared if: (1) the reset button is pressed once more (LEDs return to default state with no action selected) or (2) the USB disk is unplugged.

4. Relevant system logs can be checked to verify whether the action was completed successfully or failed. If an action is successful, LEDs will be restored to their default state when the USB disk device was first plugged in and no action was selected.
5. The USB disk device may be unplugged, after which LEDs will return to their state prior to the USB disk device being plugged in (USB LED off), and a log will be available in system logs.

On-demand Configuration backup

1. In the "Default Action" state, on-demand configuration backup to disk can be performed by holding down the reset button for more than 10 seconds. During file transfer the USB LED may blink. However, since configuration files are usually not very large, this process may finish quickly.
2. This action will save the current running configuration to the disk under the path **"/TXone/config/xxxxxx.acf"**.
3. After you have saved the config, if successful, the USB app will return to the "Default Action" state. If any error occurs, the FAULT LED will turn red.

The system logs will also reflect whether the action was successful or not.

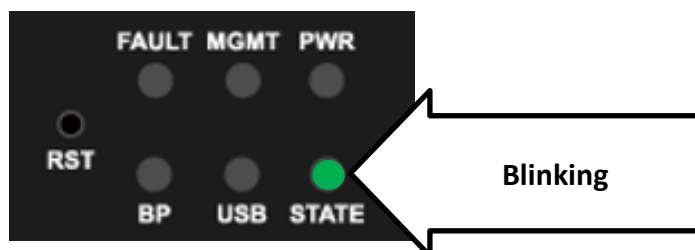
Load Pattern from Disk

A DPI pattern file may be easily and quickly loaded from a USB disk device. This functionality allows for a floor operator to update the pattern file on the physical floor of the ICS environment without the need of a client computer to log into the device.

1. Save the pattern file in a USB disk device under path **"/TXone/pattern/"**. Assuming a pattern file has the name pattern.acf, its file path on the USB disk device would be **"/TXone/pattern/pattern.acf"**.

Note: Saving pattern files under other paths or incorrect folder names will cause the file to not be detected during the pattern load process. Folder names are case-insensitive. If multiple pattern files exist in the folder, the newest will be used.

2. Plug in the USB disk. Enter the "Default Action" state.
3. In the default action state, give the reset button one short press to toggle it to its "Load Pattern" action state.
4. When the device is in its "Load Pattern" action state, the "STATE" LED indicator will change to blinking green.



5. The "Load Pattern" action must now be confirmed by holding down the reset button for more than 3 seconds.
6. After the confirmation, the selected action will be attempted. If successful, the USB app will return to the "Default Action" state. If any errors occur, the FAULT LED will turn red.
7. The system logs will also reflect whether the action was successful or not.

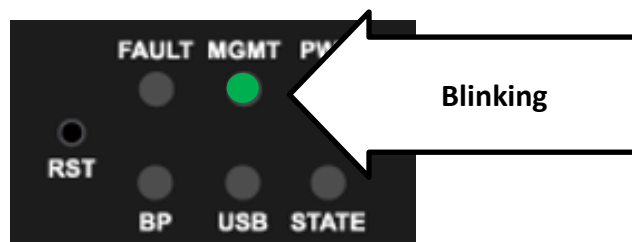
Load Configuration from Disk

A configuration file may be easily and quickly loaded via a USB disk device. This functionality allows for a floor operator to update/restore the configuration on the physical floor of the ICS environment without the need of a client computer to log into the device.

1. Save the pattern file in a USB disk device under path **"/TXone/config/"**. Assuming a pattern file has the name pattern.acf, its file path on the USB disk device would be **"/TXone/config/config.acf"**.

Note: Saving config files under other paths or incorrect folder names will cause the file to not be detected during the pattern load process. Folder names are case-insensitive. If multiple config files exist in the folder, the newest will be selected in subsequent steps

2. Plug in the USB disk. Enter the "Default Action" state.
3. In the default action state, give the reset button one short press to toggle it to its "Load Config" action state.
4. When the device is in its "Load Config" action state, the "MGMT" LED will change to blinking green.



5. The "Load Config" action must now be confirmed by holding down the reset button for more than 3 seconds.
6. After the confirmation, the action will be attempted. If successful, the USB app will return to the "Default Action" state. If any error occurs, the FAULT LED will turn red.
7. The system logs will also reflect whether the action was successful or not.

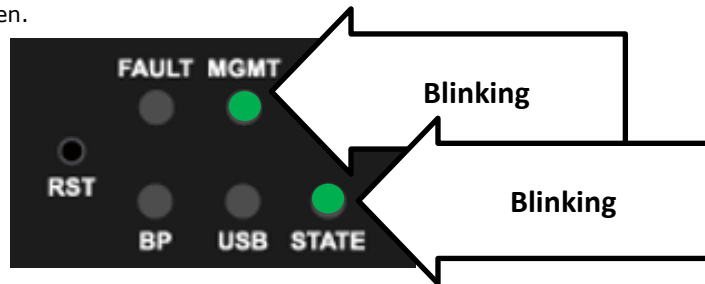
Load Firmware from Disk

Device firmware may be easily and quickly upgraded via a USB disk device. This functionality allows for a floor operator to upgrade/change the firmware of a device on the physical floor of the ICS environment without the need of a client computer to log into the device.

1. Save the pattern file in a USB disk device under path **"/TXone/firmware/"**. Assuming a pattern file has the name **pattern.acf**, its file path on the USB disk device would be **"/TXone/firmware/firmware.acf"**.

Note: Saving firmware files under other paths or incorrect folder names will cause the file to not be detected during the firmware load process. Folder names are case-insensitive. If multiple firmware files exist in the folder, the newest will be selected in subsequent steps

2. Plug in the USB disk. Enter the "Default Action" state.
3. In the default action state, give the reset button three short presses to toggle to the "Load Firmware" action state.
4. When the device is in its "Load Firmware" action state, the "MGMT" and "STATE" LEDs will change to blinking green.



5. The "Load Firmware" action must now be confirmed by holding down the reset button for more than 3 seconds.
6. After the confirmation, the action will be attempted. If successful, the USB app will return to the "Default Action" state. If any error occurs, the FAULT LED will turn red.
7. The system logs will also reflect whether the action was successful or not.

Note: Latest versions of the firmware files can be downloaded at the Trend Micro Download Center at https://www.trendmicro.com/en_us/business/products/downloads.html

Terms and Acronyms

The following table lists the terms and acronyms used in this document.

Terms/Acronyms	Definitions
ALG	Application Layer Gateway
CEF	Common Event Format
CIDR	Classless Inter-Domain Routing
DPI	Deep Packet Inspection
EWS	Engineering Workstation
HMI	Human-Machine Interface
ICS	Industrial Control System
IT	Information Technology
NAT	Network Address Translation
ODC	Operational Technology Defense Console
OT	Operational Technology
OT Defense Console	Operational Technology Defense Console
PLC	Programmable Logic Controller
SCADA	Supervisory Control and Data Acquisition